

Solving the Final Parsec Problem with Self-Interacting Dark Matter: A Theoretical Review and Numerical Verification

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Abstract

The coalescence of Supermassive Black Hole (SMBHs) binaries is an expected outcome of galaxy mergers, yet current astrophysical models fail to explain how these binaries cross the spatial gap between 10 parsecs and 0.1 parsecs. This stalling is famously known as the Final Parsec Problem. While dynamical friction from a standard Cold Dark Matter (CDM) spike has been proposed to drive the inspiral, the injected orbital energy ultimately disrupts the spike. This report reviews the recent proposal by Alonso-Álvarez et al. (2024) [3], which introduces Self-Interacting Dark Matter (SIDM) as a robust solution to the problem. Their massive mediator SIDM model naturally forms a thermalized central core that acts as a reservoir, continuously replenishing the density spike and providing sustained dynamical friction to the binary. This sustained friction predicts a distinct spectral softening in the stochastic gravitational wave background at low frequencies, providing an excellent fit to recent data from Pulsar Timing Arrays (PTAs). Finally, to independently verify the environmental mechanics of this proposed solution, this report presents an original N-body toy model using `REBOUND`, confirming that the binary’s inspiral rate is fundamentally governed by the total mass of the dark matter reservoir.

Contents

1	Introduction	3
2	The Final Parsec Problem	3
2.1	Previously proposed solutions	3
2.2	Proposed Solution	4
3	Technical Details of the proposed Solution	5
3.1	Dark Matter Spike	5
3.2	DM Distribution: NFW Profile, Core, and Spike	5
3.2.1	SIDM Core	6
3.3	SIDM with Massive Mediator	7
3.4	Computed Profiles	8
3.5	SMBH Merger Dynamics	8
3.6	Binding Energy of the DM Spike	10
3.7	Model Performance	11
4	Observable Consequence: Gravitational Waves	11
4.1	Characteristic Strain	13

5	N-Body Toy Model: Numerical Simulation of a Binary with a CDM Spike	15
5.1	Interpretation of the N-body simulations: CDM depletion and microscopic slingshots	20
5.2	Stalling distance for different spike masses	22
6	Conclusions	23
A	Derivation of the Isothermal Poisson Equation	25
A.1	Hydrostatic Equilibrium	25
A.2	The Isothermal Equation of State	25
A.3	Poisson's Equation	26
A.4	Non-Dimensionalization	26
B	Dynamical Friction: A Fluid Dynamics Approach	27
B.1	Solution in Fourier Space	28
B.2	Inverse Fourier Transform and Complex Integration	29
B.3	The Unphysical Void and Jeans Instability	29
B.4	Removing Self-Interaction	30
B.5	Return to Real Space and Causality Prescription	31
B.6	The Ultraviolet Cutoff and Wake Density	32
B.7	The Dynamical Friction Force	33
B.8	Final Conclusion: The Chandrasekhar Formula	34
C	Dynamical Friction Power Loss for a Binary System	35
C.1	D.1 Binary Kinematics	35
C.2	D.2 Total Power Dissipation	35
C.3	D.3 The Velocity Fraction $N(< v_{\text{bh}})$	36
D	Gravitational Wave Frequency and Energy Spectrum	36
D.1	1. Gravitational Wave Frequency	37
D.2	2. Orbital Energy Spectrum	37
E	Analytical Derivation of the Exponential Stalling Law	38

1 Introduction

The article focuses on how Self-Interacting Dark Matter (SIDM) can solve the Final Parsec Problem in the mergers of Supermassive Black Holes (SMBHs). There are multiple lines of evidence for these mergers, but current galactic models cannot explain their coalescence. Instead, they predict the SMBH binary will stall at a separation of approximately 1 parsec.

A dark matter spike could absorb orbital energy from the binary through dynamical friction. However, another complication arises: the binding energy of the spike cannot accommodate all the orbital energy the SMBHs need to lose. Therefore, Alonso-Álvarez et al. propose SIDM, which features a more tightly bound spike, as a mechanism to replenish the local density through thermalization. This ensures that dynamical friction continues to drive the binary together until gravitational waves become sufficiently strong.

2 The Final Parsec Problem

It is established that most galaxies in the universe possess a Supermassive Black Hole (SMBH) at their center. Because we know that galaxies frequently interact and merge, we expect their respective SMBHs to eventually merge as well.

There is compelling observational evidence for black hole coalescences, primarily through Gravitational Wave (GW) measurements. For instance, the celebrated Laser Interferometer Gravitational-Wave Observatory (LIGO) made the historic first detection of a stellar-mass black hole binary merger in 2015 (see [15]). While ground-based interferometers observe stellar-mass events, evidence for *supermassive* mergers is captured at much lower frequencies. Pulsar Timing Arrays (PTAs) have detected a stochastic GW background at nanohertz frequencies. These measurements, carried out by collaborations such as NANOGrav [2], the Parkes Pulsar Timing Array (PPTA) [11], and the European Pulsar Timing Array (EPTA) [7], are highly consistent with a cosmic background generated by merging SMBHs.

A statistical analysis conducted by NANOGrav to fit their PTA GW background data resulted in a peak merger timescale of 0.1 Gyr, with a standard deviation extending over several Gyr [1]. Thus, current observational data strongly supports the occurrence of fast SMBH mergers.

2.1 Previously proposed solutions

The first idea was that the merger is driven by interactions with surrounding stars and gas. The SMBHs can act as a gravitational slingshot, ejecting stars that pass close to them. These ejected stars carry away part of the binary's orbital energy and angular momentum, "hardening" the binary. However, it was shown that this mechanism only works down to a separation of about 10 pc due to an effect called loss cone depletion [4]. Essentially, the binary eventually runs out of local stars to scatter, causing the inspiral to stall at around 10 pc. Therefore, another driving mechanism is required at that point.

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Another proposed mechanism is the emission of GWs, which also carry away orbital energy to harden the binary. However, it can be shown that the hardening time driven purely by GW emission exceeds the age of the universe (a Hubble time) [20]. GW emission only becomes dominant and strong enough to drive the merger at roughly 0.1 pc. Hence, neither of these mechanisms is sufficient on its own, nor can they bridge the gap when

combined.

Another solution that has been suggested is dynamical friction (DF) from the accretion disks of the SMBHs. However, the most optimistic models predict that it still takes several Gyr for the merger to complete [9], or even a timescale well beyond a Hubble time [16]. Thus, circumbinary gas disks do not robustly solve the problem.

While purely spherical galaxies suffer from severe loss cone depletion [4], realistic galaxies are rarely spherical. High-resolution N-body simulations have demonstrated that in axisymmetric or triaxial galactic potentials, complex stellar orbits naturally and continuously refill the loss cone. In these environments, stellar dynamical friction remains highly efficient, allowing the SMBH binary to bridge the final parsec and merge within a Hubble time [14]. Although Khan et al. claim this resolves the Final Parsec Problem, the scientific community remains highly skeptical of their conclusions because the solution relies on a very specific galactic configuration. Furthermore, it has been shown that the violent ejection of stars by the binary actively disrupts the axisymmetry of the host galaxy, transforming the core back into a sphere and re-triggering loss cone depletion [19].

Since there is no standard astrophysical mechanism that satisfactorily drives the merger below the 1 Gyr limit, Alonso-Álvarez et al. propose that a Dark Matter (DM) spike can bridge the gap [3]. They argue that a Self-Interacting Dark Matter (SIDM) model with a massive mediator provides the necessary sustained friction to complete the merger on a sub-Gyr timescale.

2.2 Proposed Solution

The driving mechanisms of the merger can be understood through Figure 1, which illustrates the inverse relationship between the binary’s separation and the emitted gravitational wave frequency.

At distances of 10 pc or greater, interactions with stars and gas drive the merger. Conversely, at separations of 0.1 pc or less, GW emission becomes strong enough to harden the binary on a reasonable timescale.

To bridge the critical gap between 10 pc and 0.1 pc, Alonso-Álvarez et al. propose that Dark Matter drives the merger. First, they analyze dynamical friction (DF) within a standard Cold Dark Matter (CDM) framework. While they prove that DF is initially strong enough to harden the binary, the orbital energy absorbed by the CDM spike rapidly disrupts it until the spike is completely depleted—a process dynamically similar to stellar loss cone depletion. Therefore, the authors propose SIDM, which naturally forms a thermalized core within the broader DM halo. This core acts as an immense thermal reservoir that continuously replenishes the local density around the binary, providing a sustainable mechanism to solve the Final Parsec Problem.

A key observational consequence of this model is that at lower frequencies (larger orbital separations), DF absorbs the vast majority of the binary’s orbital energy. Consequently, there must be a pronounced softening, or turnover, in the GW spectrum at these low frequencies, a prediction that can be directly tested against PTA data.

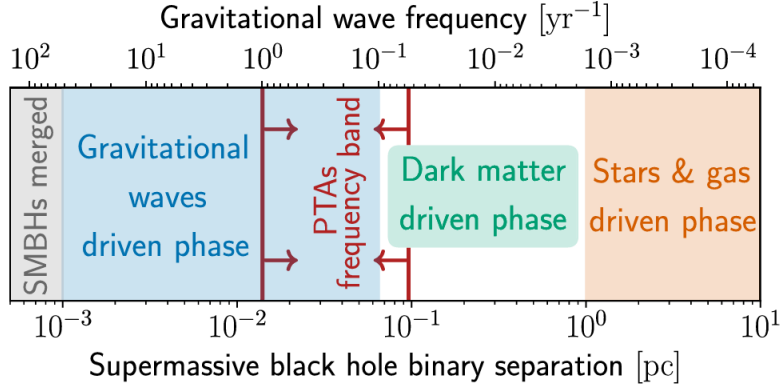


Figure 1: Dominant mechanisms driving supermassive black hole binary hardening as a function of binary separation (bottom axis) and the corresponding gravitational wave emission frequency (top axis). At large separations ($d \gtrsim 10$ pc), orbital decay is primarily driven by interactions with ambient stars and gas. At intermediate separations ($d \simeq 1$ pc), standard Cold Dark Matter spikes evaporate, leading to stalling; however, a Self-Interacting Dark Matter (SIDM) core can provide sustained dynamical friction to bridge this gap. At sub-parsec scales, energy loss is entirely dominated by gravitational wave emission, which eventually enters the Pulsar Timing Array (PTA) observational band. Figure reproduced from Alonso-Álvarez et al. (2024) [3].

3 Technical Details of the proposed Solution

3.1 Dark Matter Spike

One of the primary assumptions in their derivation is the existence of a DM spike at the center of the galaxy. There is strong theoretical evidence supporting this assumption; it was first proposed by Gondolo & Silk in 1999 [12, 30]. They mathematically proved that if a central black hole grows adiabatically within a CDM halo, the DM density profile steepens dramatically near the center, creating a massive reservoir of DM particles. An SMBH binary would therefore be immersed in a dense DM spike that generates DF, effectively hardening the binary.

3.2 DM Distribution: NFW Profile, Core, and Spike

For the CDM case, a standard Navarro-Frenk-White (NFW) density profile is assumed [21]:

$$\rho(r) = \frac{\rho_s}{\frac{r}{r_s} \left(1 + \left(\frac{r}{r_s} \right)^2 \right)} \quad (1)$$

where r_s and ρ_s are the scale radius and characteristic density, respectively. These parameters can be determined using empirical scaling relations between the DM halo mass (M_{200}) and the total mass of the SMBH binary ($M_\bullet = M_1 + M_2$) [10, 17]. Applying these relations completely determines the macroscopic halo structure; however, a localized description of the central spike is still required.

For the CDM model, the standard NFW profile must be modified at radii smaller than the spike radius ($r < r_{sp}$). The influence radius of the spike is approximately $r_{sp} \simeq 0.2r_{2M}$ [18], where r_{2M} is the radius enclosing a total DM mass equal to $2M_\bullet$. Deep inside this

region, the gravitational potential of the SMBH dominates, drawing in particles to create a dense accumulation of dark matter known as the spike.

This spike profile is governed by a power law:

$$\rho(r) = \rho_{sp} \left(\frac{r_{sp}}{r} \right)^\gamma \quad (2)$$

where ρ_{sp} is the density of the dark matter evaluated at r_{sp} .

Assuming $r_s \gg r_{2M}$, it can be analytically shown that $r_{2M}^2 = \frac{M_\bullet}{\pi r_s \rho_s}$. The steepness of the spike, γ , is treated as a free parameter, theoretically ranging from $\gamma = 1/2$ for a halo formed after a recent galactic merger, up to $\gamma = 7/3$ for the adiabatic growth of a spike.

3.2.1 SIDM Core

In the SIDM framework, the halo features a central core—a flattened density region where particle interactions are frequent enough to thermalize the inner halo. SIDM was originally proposed to solve the core-cusp problem, driven by observational data suggesting that galaxies have flatter central profiles than CDM models predict [28]. High-resolution N-body simulations have since confirmed that SIDM naturally produces these softer, cored profiles [26].

The boundary of this core is defined by the following condition:

$$(\langle \sigma v \rangle) \frac{\rho_{\text{NFW}}(r_1)}{m} t_{\text{age}} \sim 1 \quad (3)$$

This condition dictates that the expected number of scattering events per particle over the age of the halo (t_{age}) is of order unity. Solving this yields r_1 , the outer radius where the thermalized core begins ($r_1 > r > r_{sp}$).

Inside this region, the density profile is governed by the isothermal Poisson equation [5], which assumes the dark matter fluid maintains a constant temperature with an average thermal velocity v_0 (the velocity dispersion) throughout the core:

$$v_0^2 \nabla^2 \ln \rho = -4\pi G \rho \quad (4)$$

The derivation of this equation is provided in Appendix A, and its solution yields the exact density profile within the thermalized region.

Like standard CDM, SIDM also forms a density spike deep within the black hole's gravitational well. However, the radius of influence of the SMBH now depends on the thermal velocity of the core particles, v_0 [23]. This SIDM spike radius is given by:

$$r_{sp} = \frac{GM_\bullet}{v_0^2} \quad (5)$$

Crucially, the steepness of this spike (γ) is dictated by the specific particle physics of the SIDM interaction; it is no longer a free parameter.

Alonso-Álvarez et al. evaluate several different SIDM models, each characterized by a distinct velocity-dependent cross-section. This dependence is parameterized by a simple power law:

$$\sigma = \sigma_0 (v_{\text{ref}}/v_0)^a \quad (6)$$

where v_{ref} is an arbitrary reference velocity (set to 100 km/s), and σ_0 is the effective cross-section at that speed.

The power-law index a defines the underlying particle physics of the interaction. For instance, $a = 0$ represents isotropic scattering, yielding a pure contact interaction, while $a = 4$ corresponds to Rutherford-like scattering mediated by a massless force carrier.

These different interaction models fundamentally alter the expectation value of the velocity-weighted cross-section:

$$\langle \sigma v_0 \rangle = \langle \sigma_0 (v_{\text{ref}}/v_0)^a v_0 \rangle = \langle \sigma_0 v_{\text{ref}} (v_{\text{ref}}/v_0)^{a-1} \rangle \quad (7)$$

This variation directly modifies the shape of the density profile. Furthermore, as previously mentioned, the exact type of mediator dictates the steepness of the SIDM spike (from Eq. 2), as derived by Shapiro and Paschalidis [27]:

$$\gamma = (a + 3)/4 \quad (8)$$

- Contact interaction: $a = 0 \implies \gamma = 3/4$
- Massless mediator: $a = 4 \implies \gamma = 7/4$

In summary, the macroscopic CDM halo consists of a standard NFW profile that transitions directly into a central spike. These two regimes are mathematically united by imposing density continuity at the CDM transition radius r_{sp} .

In contrast, the SIDM framework is a three-part piecewise profile. It follows an NFW shape at large radii ($r > r_1$), which transitions into a thermalized core governed by the isothermal Poisson equation (Eq. 4). To solve for the interface, one must impose density continuity and require that the total enclosed mass matches the equivalent macroscopic NFW halo (M_{200}). These boundary conditions uniquely determine the thermal velocity v_0 . Finally, the central spike profile is stitched onto the system at $r < r_{sp}$ by enforcing continuity between the core and the spike.

Ultimately, the resulting dark matter spike for the SIDM case is generally shallower than its CDM counterpart. This is partially because the power-law index γ is inherently smaller, but more importantly, it is because the collisional thermalization flattens the profile at intermediate distances, creating a massive, lower-density core that cushions the binary.

3.3 SIDM with Massive Mediator

A critical SIDM model proposed by the authors is the massive mediator framework, as it yields the most successful results. This model features a broken power law, exhibiting two distinct kinematic regimes separated by a transition velocity:

$$v_t \sim c \frac{m_\gamma}{m} \quad (9)$$

where m is the mass of the dark matter particle and m_γ is the mass of the interaction mediator. Below this transition velocity ($v < v_t$), the system behaves as a pure contact interaction with $a = 0$. However, at velocities exceeding this threshold ($v > v_t$), the mediator behaves like a massless carrier with $a = 4$. The theoretical foundation for this behavior is derived by Tulin et al. [29].

A natural question arises: how can there be a regime transition if the average thermal velocity is constant throughout the entire core? The answer lies within the gravitational

influence of the spike. Deep inside the spike, the thermal velocity of the particles begins to rise as:

$$\frac{v}{v_0} \sim \frac{4}{11} \left(\frac{r_{sp}}{r} \right)^{1/2} \quad (10)$$

Therefore, as particles fall deeper into the black hole's potential well, their local velocity increases until it reaches v_t . To account for this, the authors model a two-part, piecewise spike. The outer spike region starts with a shallow profile of $\gamma = 3/4$, but once the local velocity crosses the transition threshold, the profile steepens to $\gamma = 7/4$. Because this inner region of the spike is much steeper, it retains significantly more dark matter mass, providing the robust dynamical friction needed to rapidly bridge the gap between the binary black holes.

3.4 Computed Profiles

With these considerations in place, the density profiles can be numerically computed for all the proposed models. These are represented in Figure 2. As shown, the standard CDM models (gray shaded region) reach a significantly denser spike at the absolute center. Meanwhile, the SIDM models (such as the black and blue curves) feature a distinct flattened region, which corresponds to the thermalized core.

As will be explored in the subsequent sections, solving the Final Parsec Problem requires striking a delicate physical balance. The dark matter must have enough self-interaction to thermalize the core and prevent the spike from being permanently disrupted by the binary's backreaction. However, if the interaction is too strong, the profile becomes excessively flat, causing the environment to lose its required dynamical friction (DF) strength.

3.5 SMBH Merger Dynamics

As previously discussed, at binary separations of $R \lesssim 10$ pc, the two dominant driving mechanisms for orbital hardening are the emission of gravitational waves (GWs) and dynamical friction (DF) from the central dark matter spike.

The power loss due to GW emission was derived by Peters in 1964 [24] and is given by the following expression:

$$P_{gw} = \frac{32}{5} q^2 (1+q) \frac{G^4}{c^5} \left(\frac{M_1}{R} \right)^5 \quad (11)$$

where $q = M_2/M_1$ is the mass ratio, with $M_1 \geq M_2$. Notice in Eq. (11) that the power loss is proportional to R^{-5} ; it increases sharply as the black holes approach one another, but is overwhelmingly weak at large radii.

Meanwhile, the power loss due to DF with the DM spike follows from the Chandrasekhar dynamical friction equation [6]. This equation can also be derived entirely from a fluid dynamics framework, as demonstrated in Appendix B, yielding the expression:

$$P_{df} = 12\pi q^2 \sqrt{1+q} (GM_1)^{3/2} R^{1/2} \times \left[\frac{N_1(q)}{q^3} \rho_{sp} \left(\frac{qR}{1+q} \right) + N_2(q) \rho_{sp} \left(\frac{R}{1+q} \right) \right] \quad (12)$$

where $N_1 = N_2 \approx 1$ for standard CDM, and $N_1 \simeq N_2 \simeq 0.2$ for SIDM.

The physical nature of the dark matter dictates the magnitude of these drag coefficients. For standard CDM, the environment is kinematically "cold," meaning the binary's

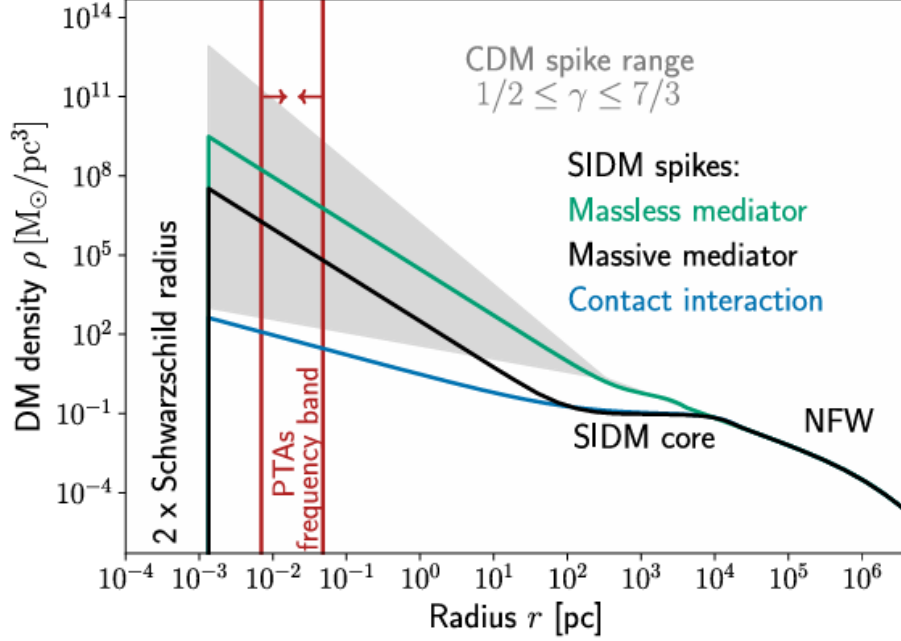


Figure 2: Dark matter density profiles (ρ) as a function of radius (r) from the central supermassive black hole. The gray shaded region represents the expected range for standard Cold Dark Matter (CDM) spikes, which are characterized by steep power-law indices ($1/2 \leq \gamma \leq 7/3$). While Self-Interacting Dark Matter (SIDM) models successfully form a constant-density core at larger radii ($r \sim 10^2 - 10^4$ pc) due to thermalization, the density re-steepens into a spike deep within the black hole’s gravitational potential. The colored lines illustrate different SIDM particle physics models: a massless mediator (green), a massive mediator (black), and a pure contact interaction (blue). The vertical red lines denote the binary separations corresponding to the Pulsar Timing Array (PTA) emission band. The massive mediator SIDM model provides the optimal density required to drive the binary through the final parsec without over-suppressing the observable gravitational wave signal. Figure reproduced from Alonso-Álvarez et al. (2024) [3].

orbital velocity greatly exceeds the thermal velocity dispersion of the background particles. This supersonic regime maximizes drag efficiency ($N \approx 1$). In contrast, SIDM forms a thermalized, kinematically "hot" core characterized by a high velocity dispersion. Because the binary is orbiting subsonically relative to this agitated dark matter fluid, the drag efficiency is inherently reduced, calibrating the coefficients to $N \simeq 0.2$.

By conservation of energy, the total orbital power loss is the sum of these dissipative mechanisms:

$$P_{\text{gw}} + P_{\text{df}} = P_{\text{orb}} \quad (13)$$

where P_{orb} is easily obtained by differentiating the orbital energy of the binary with respect to time:

$$P_{\text{orb}} = |\dot{E}_{\text{orb}}| = qGM_1^2 \frac{|\dot{R}|}{2R^2} \quad (14)$$

Substituting the three power expressions, we arrive at a first-order differential equation describing the orbital separation between the SMBHs:

$$|\dot{R}| = \frac{2R^2}{qGM_1^2} (P_{\text{gw}} + P_{\text{df}}) \quad (15)$$

Numerical integration of this equation tracks the evolution of the orbital separation and determines the total inspiral time for the various dark matter models.

3.6 Binding Energy of the DM Spike

Numerical integration of Eq. 15 shows that the inspiral time for a standard CDM spike would be less than 1 Gyr. If we blindly assumed this kinematic result were the complete picture, the Final Parsec Problem would be solved. However, we must also determine whether the DM spike can physically absorb all the orbital energy injected by the SMBH binary without being destroyed.

First, we compute the total change in orbital energy as the binary shrinks from 10 pc down to 0.1 pc:

$$\Delta E_{\text{orb}} = \frac{qGM_1^2(R_{\text{gw}}^{-1} - R_*^{-1})}{2} \quad (16)$$

where $R_{\text{gw}} \simeq 0.1$ pc is the radius at which GW emission becomes sufficiently strong to drive the merger, and $R_* \simeq 10$ pc is the distance at which stellar interactions cease to be effective due to loss cone depletion. Next, the gravitational binding energy of the spike must be evaluated. Its analytical expression is:

$$\frac{U}{G} = M_{\bullet} \int d^3x \frac{\rho_{\text{sp}}(\vec{x})}{|\vec{x}|} + \int d^3x_1 d^3x_2 \frac{\rho(\vec{x}_1)\rho(\vec{x}_2)}{|\vec{x}_1 - \vec{x}_2|} = 4\pi M_{\bullet} \rho_{\text{sp}} r_{\text{sp}}^2 \left(\frac{1 - \epsilon^{2-\gamma_{\text{sp}}}}{2 - \gamma_{\text{sp}}} \right) + 4\pi^2 \rho_{\text{sp}}^2 r_{\text{sp}}^5 g(\gamma) \quad (17)$$

where ϵ is the ratio of the Schwarzschild radius to the spike radius, and $g(\gamma)$ is a geometric factor where $g(\gamma \leq 2) \sim 3$, growing to $g(7/3) \sim 20$.

Both terms in Equation 17 are of the same order. Crucially, for a standard CDM spike, this total binding energy is roughly 4 to 5 orders of magnitude smaller than the required orbital energy dissipation (ΔE_{orb}). Therefore, the spike cannot absorb the orbital energy without being completely disrupted.

Consequently, collisionless CDM cannot solve the Final Parsec Problem, a limitation previously demonstrated by Kavanagh et al. [13]. SIDM, however, overcomes this barrier. In physical terms, collisional SIDM behaves like a compressible gas; as the immediate spike

is heated and disrupted by the binary, the vast, thermalized outer core expands inward to continuously replenish the spike. This fluid-like behavior allows the SIDM core to act as an infinite heat sink, successfully absorbing the orbital energy without permanent depletion.

3.7 Model Performance

In summary, we impose two critical physical constraints on the system:

- The inspiral time must be less than 1 Gyr.
- The binding energy of the core must be greater than the injected orbital energy (ΔE_{orb}). We evaluate the binding energy of the core rather than just the spike because the core is actively thermalized by particle interactions, acting as a vast energy reservoir to replenish the spike.

Both the inspiral time and the binding energy can be computed numerically. The inspiral time is evaluated via Eq. (15), and the binding energy of the core is approximated by:

$$U_{\text{core}}^{\text{SIDM}} \simeq \frac{1}{2} M_{\text{core}} v_0^2 \quad (18)$$

where M_{core} is the total mass enclosed within the core, which depends implicitly on the interaction cross-section and the specific particle physics model. The thermal velocity v_0 exhibits a similar parameter dependence.

This brings us to the first major results from Alonso-Álvarez et al. [3].

In Figure 3, the various SIDM models are plotted in different colors. The top panel displays the inspiral time as a function of the effective cross-section (σ_0/m). The unshaded white region indicates the parameter space that successfully satisfies the $t_{\text{df}} < 1$ Gyr condition. Every model except the pure contact interaction ($a = 0$) satisfies this condition over at least some portion of the cross-section range.

The bottom panel illustrates the core binding energy constraint for the same values of σ_0/m . Again, the unshaded white region highlights where the condition $U_{\text{core}}^{\text{SIDM}} > \Delta E_{\text{orb}}$ is satisfied. This physical constraint is significantly more restrictive, with only three models successfully surviving the energy injection.

The theoretical trade-off is clear: a larger interaction cross-section increases the binding energy and resilience of the core, but it simultaneously flattens the spike, which severely weakens the dynamical friction and slows the inspiral. Crucially, the only model that simultaneously satisfies both constraints across a viable parameter window is the massive mediator framework (solid black line).

4 Observable Consequence: Gravitational Waves

As previously suggested, dynamical friction will fundamentally alter the emitted GW spectrum. Because a significant fraction of the orbital energy is dissipated via DF rather than GW emission at low frequencies (large orbital separations), this produces a characteristic "softening," or attenuation, of the spectrum in that regime.

For a circular binary, the emitted GW frequency (f_s) is exactly twice the orbital frequency:

$$f_s = \frac{\omega}{\pi} = \frac{1}{\pi} \left(\frac{GM_{\bullet}}{R^3} \right)^{1/2} \quad (19)$$

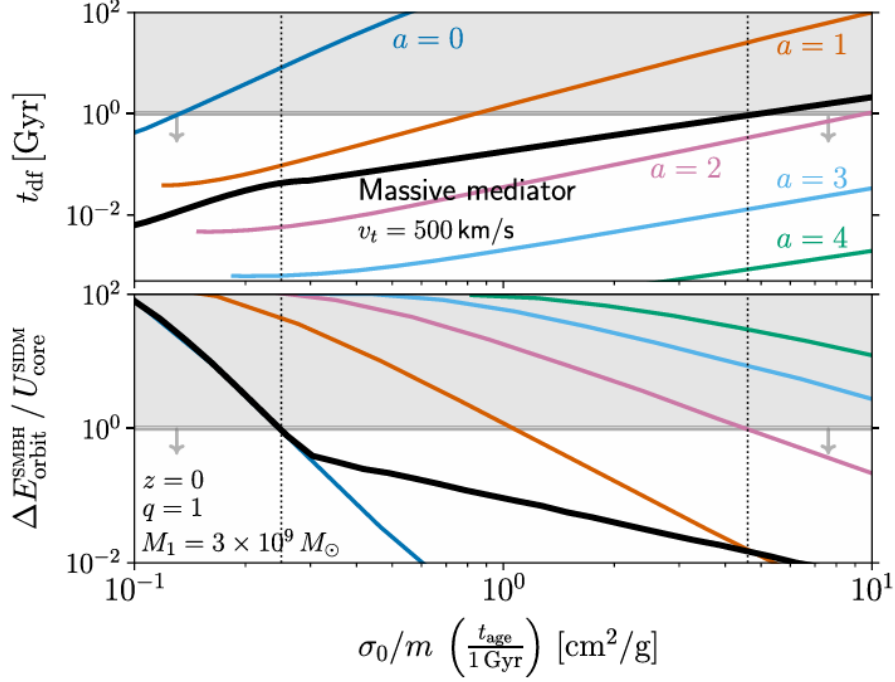


Figure 3: Dynamical friction timescale and core depletion constraints as a function of the effective Self-Interacting Dark Matter (SIDM) cross-section. **Top panel:** The timescale (t_{df}) required for the supermassive black hole binary to harden through the final parsec. The gray shaded region indicates models where the inspiral takes longer than 1 Gyr, failing to merge efficiently within a typical galaxy merger timeline. **Bottom panel:** The ratio of the orbital energy dissipated by the binary (ΔE_{orbit}^{SMBH}) to the total gravitational binding energy of the SIDM core (U_{core}^{SIDM}). The shaded region indicates where the injected energy exceeds the binding energy, resulting in the catastrophic dispersal of the dark matter core and the subsequent stalling of the binary. The colored lines represent phenomenological velocity-dependent scattering models scaling as v^{-a} . The black line represents the physically motivated massive mediator model. The vertical dotted lines bound the "Goldilocks" viable parameter space: the exact cross-section window where the massive mediator provides enough friction to merge the binary rapidly (top), but not so much friction that it destroys the dark matter core (bottom). Figure reproduced from Alonso-Álvarez et al. (2024) [3].

This can be conveniently expressed in terms of the system's dimensionless units (see Appendix D):

$$f_s = \frac{1}{2\pi t_{\text{sp}}} \sqrt{\frac{1+q}{2x^3}} \quad (20)$$

To predict the observable signal, the key quantity to evaluate is the energy radiated per unit frequency, which is also derived in Appendix D:

$$\frac{dE_{\text{gw}}}{df_s} = \frac{qGM_1^2}{6f_s x r_{\text{sp}}} \left(\frac{P_{\text{gw}}}{P_{\text{gw}} + P_{\text{df}}} \right) \quad (21)$$

Because there are two competing mechanisms draining the binary's orbital energy, we simply differentiate the total orbital energy with respect to frequency, and then multiply the result by the branching ratio—the fraction of power radiated purely as gravitational waves, $\frac{P_{\text{gw}}}{P_{\text{gw}} + P_{\text{df}}}$.

4.1 Characteristic Strain

Thus far, we have studied the gravitational wave emission of a single isolated binary. However, to study the stochastic GW Background (GWB), we must account for the aggregate emission of all merging binaries across the universe.

This is quantified by the characteristic strain, h_c , which describes the amplitude of spacetime "deformation" at each observable frequency. This quantity is inherently proportional to the spectral energy emitted per frequency bin, $\frac{dE_{\text{gw}}}{df_s}$. By integrating this energy over the entire cosmological population of observable binaries—specifically integrating over redshift (z), total binary mass (M), and mass ratio (q)—we obtain:

$$h_c^2(f) = \frac{4G}{\pi c^2 f} \int dz dM dq \frac{d^3n}{dz dM dq} \frac{dE_{\text{gw}}}{df_s} \quad (22)$$

The synthetic population of SMBH binaries employed in this analysis was extracted from Chen et al. [8], utilizing the same demographic models adopted by the NANOGrav collaboration [1]. In Eq. (22), f_s denotes the frequency at the source, which is related to the observed Earth frequency f by the cosmological redshift: $f_s = f(1+z)$. By combining this population model with the energy spectrum $\frac{dE_{\text{gw}}}{df_s}$ for the massive mediator SIDM framework (the only model satisfying both the kinematic and thermodynamic constraints), we can compute the predicted characteristic strain and compare it directly with observational PTA data.

Figure 4 presents the characteristic strain as a function of observed frequency. The discrete data points represent measurements from NANOGrav (gray) [1], the PPTA (orange) [11], and the EPTA (green) [7], all shown with their respective uncertainty bars.

Superimposed on this data are the theoretical fits for different environmental models. The dashed red line represents a binary population evolving purely via GW emission with no dark matter; this model fails to soften at low frequencies and physically suffers from the Final Parsec Problem. The dashed blue line models a standard CDM environment, which mathematically fits the low-frequency turnover but completely fails to account for the destructive backreaction of the spike (as collisionless CDM cannot survive the injection of orbital energy).

Finally, the solid black line represents the massive mediator SIDM model with an effective cross-section of $\sigma_0/m = 6 \text{ cm}^2/\text{g}$ and a transition velocity of $v_t = 500 \text{ km/s}$. This

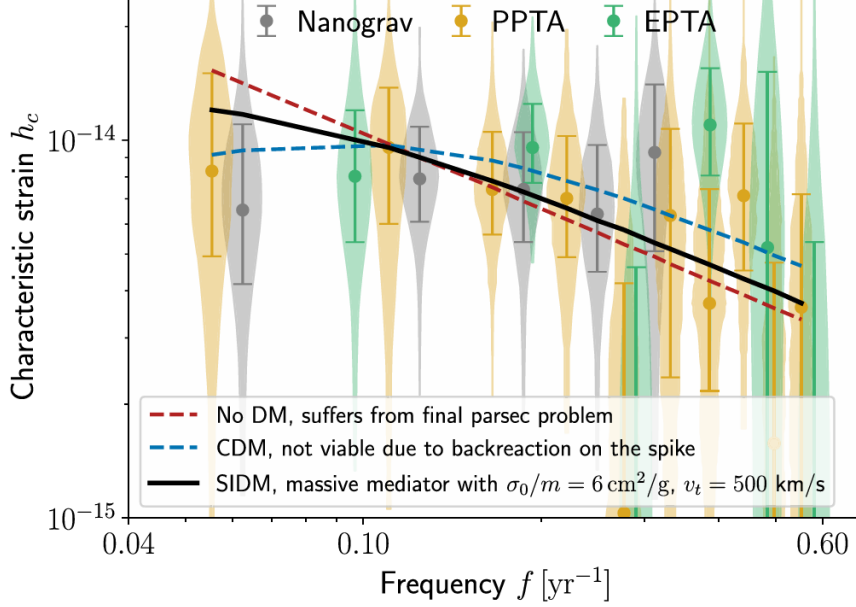


Figure 4: Characteristic strain (h_c) of the stochastic gravitational wave background as a function of frequency. Observational data points and corresponding uncertainty distributions (violin plots) are shown for three major Pulsar Timing Array collaborations: NANOGrav (gray), the Parkes Pulsar Timing Array (PPTA, yellow), and the European Pulsar Timing Array (EPTA, green). The theoretical models illustrate the impact of different binary evolutionary environments. The red dashed line represents a purely gravitational-wave-driven population (yielding a standard $f^{-2/3}$ spectrum); while it appears to fit the higher frequencies, it physically fails to merge in a Hubble time (the Final Parsec Problem). The blue dashed line models a standard Cold Dark Matter (CDM) environment, which excessively attenuates the low-frequency signal and is dynamically unviable due to the rapid disruption of the density spike. The solid black line demonstrates the optimal Self-Interacting Dark Matter (SIDM) model employing a massive mediator ($\sigma_0/m = 6 \text{ cm}^2/\text{g}$, $v_t = 500 \text{ km/s}$). This model provides sufficient dynamical friction to successfully drive the binaries to merger, while naturally producing a spectral flattening at low frequencies that is in excellent agreement with the global PTA datasets. Figure reproduced from Alonso-Álvarez et al. (2024) [3].

model excellently fits the observed data points by correctly predicting the low-frequency spectral softening (since orbital energy is efficiently dissipated via DF in this regime). Crucially, this model physically withstands the orbital energy injection without destroying the core, providing a complete and robust solution to the Final Parsec Problem.

Statistical Methodology and Parameter Constraints

To robustly constrain the viable parameter space for the Self-Interacting Dark Matter (SIDM) model, Alonso-Álvarez et al. (2024) employ a rigorous χ^2 goodness-of-fit analysis [3]. The fundamental objective of this statistical treatment is to quantify how accurately the theoretical characteristic strain spectrum, generated by a specific combination of dark matter parameters, reproduces the observational data.

For any given dark matter halo velocity dispersion (v_t) and effective scattering cross-section (σ_0/m), the model predicts a specific theoretical strain, $h_{c,\text{th}}(f)$. This prediction is evaluated against the discrete strain measurements, $h_{c,\text{obs}}(f_i)$, reported by the global Pulsar Timing Array (PTA) collaborations across individual frequency bins. The χ^2 statistic is constructed by summing the squared deviations between the theoretical and observed values, inversely weighted by the observational uncertainty (variance) at each frequency bin, $\sigma_{\text{obs},i}$:

$$\chi^2(v_t, \sigma_0/m) = \sum_i \left(\frac{h_{c,\text{obs}}(f_i) - h_{c,\text{th}}(f_i; v_t, \sigma_0/m)}{\sigma_{\text{obs},i}} \right)^2 \quad (23)$$

By computing Equation 23 across a dense grid of particle physics parameters, one generates a topological map of the fit quality, as shown in Figure 5. The χ^2 contours explicitly delineate the statistically permitted regions of the parameter space at various confidence intervals. The global minimum of this surface ($\chi^2 \approx 15$) identifies the optimal best-fit model. Consequently, this statistical minimization mathematically proves that a massive mediator SIDM configuration naturally and successfully reproduces the low-frequency spectral turnover necessitated by the combined NANOGrav, EPTA, and PPTA datasets, while simultaneously satisfying the dynamical constraints of the core binding energy.

5 N-Body Toy Model: Numerical Simulation of a Binary with a CDM Spike

To observe how standard CDM affects the orbit of a binary, I ran an N-body simulation using the REBOUND package in Python [25]. The simulation utilizes Newtonian dynamics, which are sufficient to capture the macroscopic effects of dynamical friction. Because the code models point particles, we regularize the gravitational interactions using a softening parameter b , which we set equal to the Schwarzschild radius of the primary SMBH:

$$r_s = \frac{2GM}{c^2} \quad (24)$$

Additionally, we impose an absorbing boundary condition for close encounters: each time a simulated DM particle approaches within a distance r_s , the software permanently merges it with the SMBH.

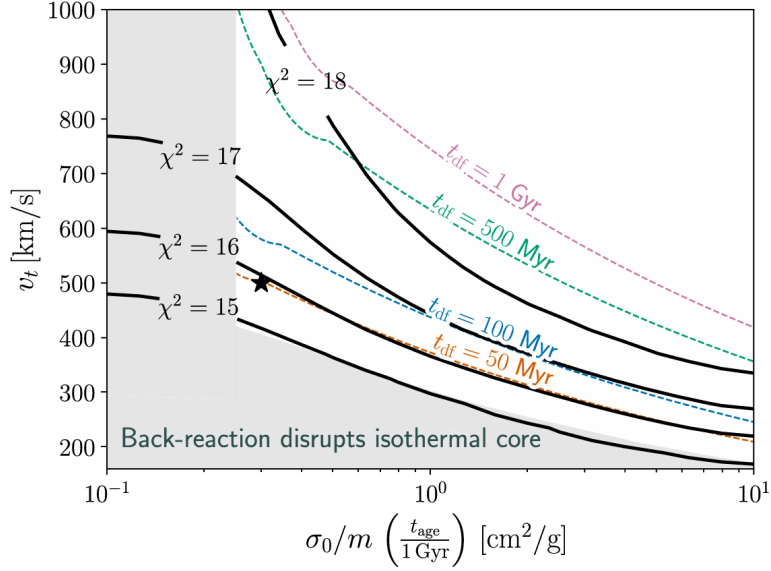


Figure 5: Parameter space constraints and global statistical fit for the massive mediator Self-Interacting Dark Matter (SIDM) model. The plot maps the halo’s thermal velocity dispersion (v_t) against the effective scattering cross-section (σ_0/m). Solid black lines indicate contours of constant χ^2 , representing the goodness-of-fit between the model’s predicted gravitational wave spectrum and the combined Pulsar Timing Array datasets; the black star marks the global best-fit parameter combination ($v_t \approx 500$ km/s and $\sigma_0/m \approx 0.3$ cm²/g scaled by age, corresponding to the reference values used in prior figures). Dashed colored lines trace contours of constant dynamical friction timescale (t_{df}), explicitly showing how quickly the binary bridges the final parsec. The gray shaded region at the bottom bounds the unviable parameter space where the binary’s orbital energy injection (back-reaction) exceeds the binding energy of the system, catastrophically disrupting the isothermal SIDM core. The unshaded central wedge represents the viable "Goldilocks" parameter space where the dark matter core is massive enough to survive the inspiral while providing optimal friction to fit the observations. Figure reproduced from Alonso-Álvarez et al. (2024) [3].

We first executed this simulation for a binary within a spike of fixed total mass ($M_{\text{spike}} = 0.1M_1$), while sweeping across different values for the total number of DM simulation particles (N_{DM}). The goal was to verify if the orbital decay depends artificially on the numerical resolution. If the result is independent of N_{DM} , it proves our simulation is numerically robust, allowing us to run subsequent models with a computationally reasonable number of particles and extrapolate the physical conclusions safely.

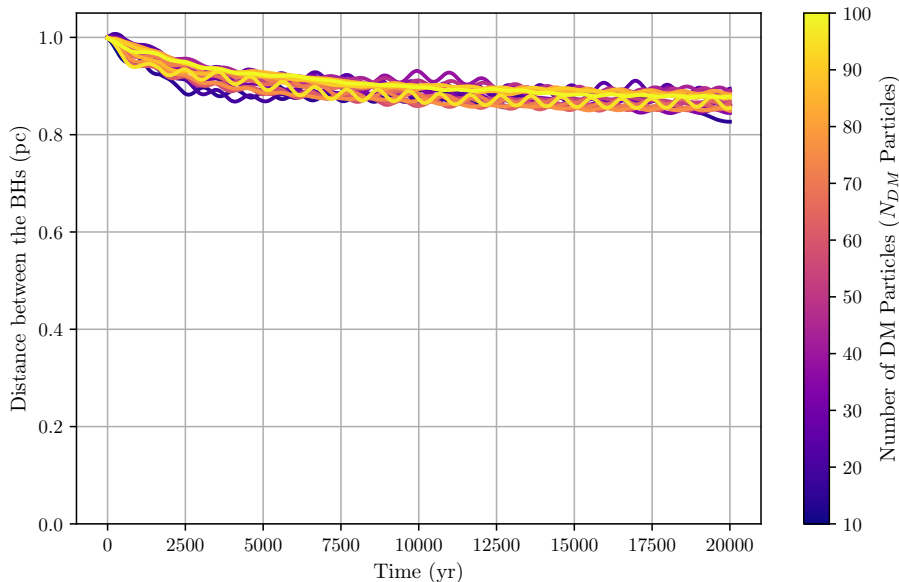


Figure 6: Evolution of binary black hole separation in a dark matter spike, simulated using the **REBOUND** code. The figure shows the time-dependent distance between the primary and secondary black holes (in parsecs) for a range of simulation particle counts, from $N_{\text{DM}} = 10$ (purple) to $N_{\text{DM}} = 100$ (yellow). In all cases, the total mass of the dark matter spike is held fixed at 0.1 times the mass of the primary black hole ($0.1M_1$).

In Figure 6, we plot the distance between the SMBHs (starting from an initial separation of 1 pc) versus time for the different values of N_{DM} . Note that all the curves exhibit a very similar trajectory; they decay at the same rate until they reach an identical plateau where the system stalls. This convergence indicates that the defining physical characteristic governing the friction is the total mass of the DM spike, not the discrete mass of the individual particles that compose it. In our simulation, the binary hardens primarily due to gravitational slingshot interactions. Once there is no more dark matter mass left in the local spike to eject, the system stalls, perfectly replicating the Final Parsec Problem.

The next logical step is to analyze the system’s behavior when varying the total mass of the dark matter spike. Figure 7 displays the orbital evolution curves for different spike masses (M_{sp}) while maintaining a fixed number of simulation particles (N_{DM}).

As expected, the rate of orbital hardening is directly proportional to the spike mass; heavier spikes induce a significantly faster inspiral. These results align perfectly with theoretical predictions: a more massive spike provides a larger reservoir of dark matter particles for the binary to eject via gravitational slingshots, thereby extracting more orbital energy and hardening the binary more efficiently.

Future analyses within this numerical framework could investigate the exact critical spike mass required to successfully drive the binary to a complete merger. Furthermore, it

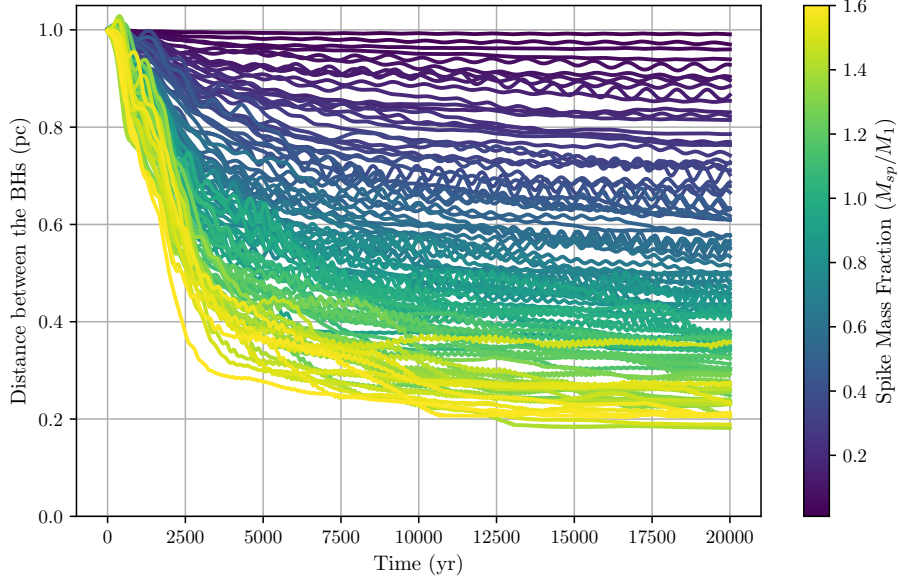


Figure 7: Evolution of the binary black hole orbital separation for varying dark matter spike masses, simulated using the `REBOUND` N-body code. The figure illustrates the time-dependent distance between the primary and secondary black holes (in parsecs) over a 20,000-year integration. To isolate the physical effect of the environmental mass, the numerical resolution is held constant at $N_{\text{DM}} = 80$ particles across all runs. The color bar indicates the total mass of the dark matter spike, expressed as a fraction of the primary black hole’s mass (M_{sp}/M_1). The trajectories clearly demonstrate the direct dependence of dynamical friction on the background density: heavier dark matter spikes (yellow/green lines) exert a significantly stronger drag force, efficiently extracting orbital energy and driving a rapid inspiral. Conversely, low-mass spikes (dark purple lines) lack the necessary gravitational drag, resulting in minimal orbital decay and effectively stalling the binary. This highlights the absolute necessity of a massive, resilient dark matter reservoir to successfully bridge the final parsec.

would be highly informative to quantify the mathematical dependence of the final plateau distance (the stalling radius) on the initial mass of the spike.

Thus far, all simulations have assumed an equal-mass binary ($q = 1$). To complete the numerical analysis, we perform a final suite of simulations where the spike mass (M_{sp}) and particle resolution (N_{DM}) are strictly fixed, while the SMBH mass ratio ($q \equiv M_2/M_1$) is systematically varied from 0.2 to 1.0.

The results are presented in Figure 8. Observe that highly asymmetric binaries (smaller q values) experience a significantly steeper and more rapid orbital decay. This accelerated hardening can be understood through the relative inertia of the system: because the total mass of the binary is lower for smaller q values (assuming M_1 remains fixed), the constant-mass dark matter spike is comparatively more effective at draining the system’s orbital energy. This perfectly complements the dynamic demonstrated in Figure 7, confirming that the inspiral rate is fundamentally governed by the relative mass ratio between the dark matter reservoir and the binary itself.

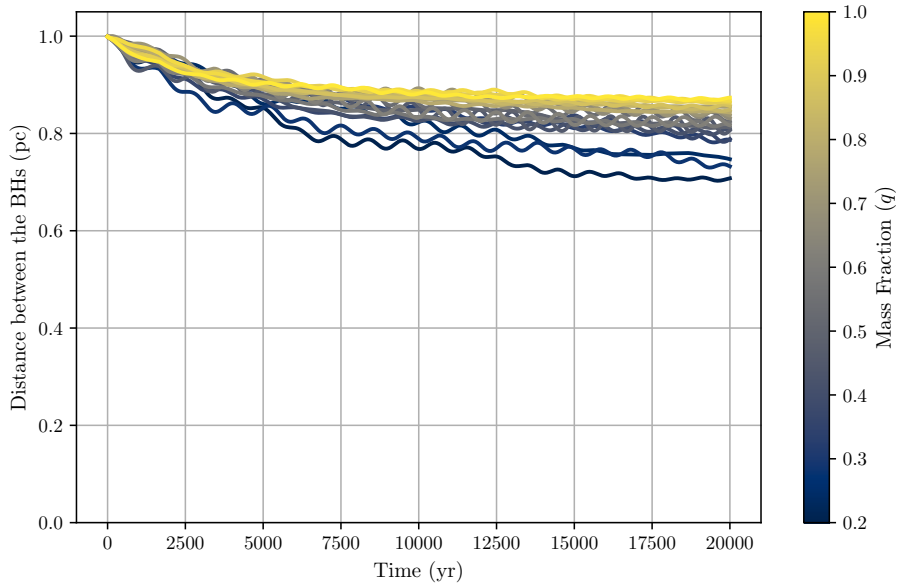


Figure 8: Evolution of the binary black hole orbital separation for varying binary mass ratios (q), simulated using the **REBOUND** N-body code. The figure tracks the time-dependent distance between the primary and secondary black holes (in parsecs) over a 20,000-year integration. To isolate the dynamical effects of the binary’s mass configuration, the dark matter spike mass and the particle resolution are held constant across all runs. The color bar denotes the mass ratio $q \equiv M_2/M_1$, ranging from highly asymmetric binaries ($q = 0.2$, dark blue) to equal-mass binaries ($q = 1.0$, yellow). The trajectories reveal that asymmetric binaries exhibit a steeper, more efficient inspiral compared to their equal-mass counterparts. This accelerated decay for lower q values is consistent with the analytical formulation of dynamical friction power loss in a centered density spike (e.g., Equation 12), where the combined drag accounting for both the primary black hole’s reflex motion near the densest center of the spike and the secondary’s extended orbit results in a higher relative energy dissipation rate for unequal masses.

5.1 Interpretation of the N-body simulations: CDM depletion and microscopic slingshots

The purpose of the REBOUND simulations presented above was not to model the full collisional SIDM scenario. Instead, the goal was to isolate the collisionless CDM limit and verify the basic mechanism responsible for the failure of a finite dark matter reservoir. The simulations therefore intentionally use a Newtonian, collisionless, discrete-particle description. In this regime, the binary does not interact with the environment as a continuous fluid, but rather through individual gravitational encounters with dark matter particles.

This is most naturally interpreted in the original Chandrasekhar picture of dynamical friction. In Chandrasekhar’s derivation, a massive body experiences a sequence of fluctuating gravitational forces produced by nearby particles. The microscopic force is stochastic, but its first moment produces a systematic deceleration, while its second moment produces diffusion in velocity space [6]. In the present binary problem, the microscopic encounters are not merely weak two-body deflections: many of the relevant events are three-body gravitational slingshots involving the dark matter particle and both black holes. Nevertheless, the logic is the same. Individual encounters provide discrete gravitational kicks, and the coarse-grained average of many such kicks appears as an effective dynamical-friction force.

This interpretation is preferable to a purely gaseous wake picture for the present simulation. The fluid formulation of dynamical friction, as developed for a perturber moving through a gaseous medium by Ostriker [22], describes the drag as the gravitational attraction between the perturber and a hydrodynamic density wake. That picture is appropriate when the background behaves as a collisional fluid. By contrast, the REBOUND calculation follows individual collisionless particles. Therefore, the relevant microscopic process is better viewed as Chandrasekhar-like stochastic gravitational scattering, with slingshot encounters providing the discrete events whose average produces a macroscopic drag.

At the microscopic level, a dark matter particle enters the interaction region of the binary with an incoming velocity $\mathbf{v}_{\chi,\text{in}}$ and leaves with an outgoing velocity $\mathbf{v}_{\chi,\text{out}}$ after interacting gravitationally with both black holes. The total change in the particle momentum can be decomposed as

$$\Delta\mathbf{p}_{\chi} = \Delta\mathbf{p}_{\chi,1} + \Delta\mathbf{p}_{\chi,2},$$

where $\Delta\mathbf{p}_{\chi,i}$ denotes the impulse received by the dark matter particle from the i -th black hole. By Newton’s third law, the corresponding impulse exerted on that black hole is

$$\Delta\mathbf{P}_i = -\Delta\mathbf{p}_{\chi,i}.$$

After many such encounters, the discrete impulses can be coarse-grained into an effective force on each black hole:

$$\mathbf{F}_i^{\text{eff}} = \frac{d\mathbf{P}_i}{dt} = - \int d\Gamma \langle \Delta\mathbf{p}_{\chi,i} \rangle.$$

Here $d\Gamma$ is the differential encounter rate, and the brackets denote an average over impact parameters, binary orbital phase, and the incoming dark matter velocity distribution. In a local scattering picture around the i -th black hole, the encounter rate may be written schematically as

$$d\Gamma_i = f(\mathbf{x}_i, \mathbf{v}_{\chi}) u_{i,\text{in}} d\sigma_i d^3v_{\chi},$$

where $f(\mathbf{x}_i, \mathbf{v}_\chi)$ is the dark matter phase-space distribution evaluated at the position of the i -th black hole, $d\sigma_i$ is the differential gravitational scattering cross section, and

$$\mathbf{u}_{i,\text{in}} = \mathbf{v}_{\chi,\text{in}} - \mathbf{V}_i$$

is the incoming relative velocity between the dark matter particle and the i -th black hole. The relative velocity entering the encounter flux is therefore the pre-scattering relative velocity.

This effective force should not be interpreted only as a net force on the binary center of mass. For binary hardening, the relevant quantity is the work done on the internal orbital motion. The corresponding orbital power is

$$\frac{dE_{\text{bin}}}{dt} = \sum_{i=1}^2 \mathbf{F}_i^{\text{eff}} \cdot \mathbf{V}_i.$$

For a circular equal-mass binary in the center-of-mass frame,

$$\mathbf{V}_1 = -\mathbf{V}_2.$$

Thus, the net force on the binary center of mass can vanish,

$$\mathbf{F}_1^{\text{eff}} + \mathbf{F}_2^{\text{eff}} \simeq 0,$$

while the internal orbital energy still decreases:

$$\mathbf{F}_1^{\text{eff}} \cdot \mathbf{V}_1 + \mathbf{F}_2^{\text{eff}} \cdot \mathbf{V}_2 < 0.$$

This is the crucial point: hardening does not require a net acceleration of the binary center of mass. It requires a tangential drag on each black hole, opposite to its own orbital motion.

Equivalently, the same process can be described from the perspective of the dark matter particles. A gravitational slingshot event transfers orbital energy from the binary to a particle. A particle enters the interaction region, interacts gravitationally with one or both black holes, and is ejected with a larger kinetic energy:

$$\Delta E_\chi = \frac{1}{2}m_\chi v_{\chi,\text{out}}^2 - \frac{1}{2}m_\chi v_{\chi,\text{in}}^2 > 0.$$

By conservation of energy, this gain must be balanced by a loss of orbital energy from the binary:

$$\Delta E_{\text{bin}} = -\Delta E_\chi.$$

Since the Newtonian binding energy of the binary is

$$E_{\text{bin}} = -\frac{GM_1M_2}{2R},$$

a negative change in E_{bin} makes the binary more tightly bound and decreases the separation R . This is the microscopic gravitational slingshot picture: particles are kicked out of the central region, while the binary hardens.

Averaging over many scattering events gives

$$\frac{dE_{\text{bin}}}{dt} = - \int d\Gamma \langle \Delta E_\chi \rangle = \sum_{i=1}^2 \mathbf{F}_i^{\text{eff}} \cdot \mathbf{V}_i.$$

This equation makes explicit the connection between the microscopic and macroscopic descriptions. The left-hand side describes the hardening of the binary. The first expression on the right-hand side describes the energy carried away by the ejected dark matter particles. The second expression describes the same effect as an effective drag force on each black hole. Therefore,

$$\text{many microscopic slingshots} \implies \text{effective dynamical friction.}$$

In this sense, the slingshot mechanism is not separate from dynamical friction; rather, it is one possible microscopic realization of the stochastic gravitational encounters whose averaged effect is described as dynamical friction.

In macroscopic language, the dark matter spike is heated by the binary. The binary loses orbital energy, while the surrounding dark matter gains kinetic energy, expands, and partially evacuates the central region. This is the backreaction of the spike. In a collisionless CDM spike, the same slingshot events that provide the initial dynamical friction also remove the particles that generate it. The local phase-space reservoir is therefore progressively depleted. Once the available particles have been heated or ejected, the effective dynamical friction weakens and the binary stalls. This is the dark-matter analogue of stellar loss-cone depletion.

The SIDM proposal should be understood as a remedy to this depletion, not as the regime directly simulated here. In SIDM, self-interactions transport heat and particles through the halo, allowing the thermalized core to replenish the central spike after it has been heated by the binary. Thus, the microscopic gravitational scattering mechanism remains the channel through which the binary transfers energy to the environment, but the collisional nature of SIDM prevents the local reservoir from being permanently depleted. The contrast between the collisionless behavior observed in the simulations and the sustained replenishment expected in SIDM is precisely the physical motivation for the SIDM solution to the final parsec problem.

5.2 Stalling distance for different spike masses

One of the main features of the collisionless simulations is that, after an initial phase of orbital decay, the binary eventually stalls. This occurs because the finite dark matter spike is progressively heated and depleted by the binary. Once the local reservoir of particles available for gravitational slingshot interactions has been exhausted, the effective dynamical friction weakens and the binary separation approaches an approximately constant value.

It is therefore useful to study how this final stalling distance depends on the total mass of the dark matter spike. This provides a simple way to estimate whether a sufficiently massive CDM spike could, in principle, drive the binary down to the separation at which gravitational wave emission becomes dominant. If the required spike mass could be constrained observationally, this relation would also provide a direct diagnostic of whether a collisionless dark matter environment is capable of bridging the Final Parsec gap.

As derived in Appendix E, a simple energy-exchange argument predicts an approximately exponential dependence of the stalling distance on the spike mass. In the idealized case, one expects

$$R_{\text{stall}} \simeq R_{\text{initial}} \exp \left[-C \left(\frac{M_{\text{spike}}}{M_1} \right) \right],$$

where C is an effective dimensionless efficiency parameter that encodes the average energy extracted per unit ejected dark matter mass. Because the simulated spike is not a uniform

Final Stalling Distance vs. Dark Matter Spike Mass

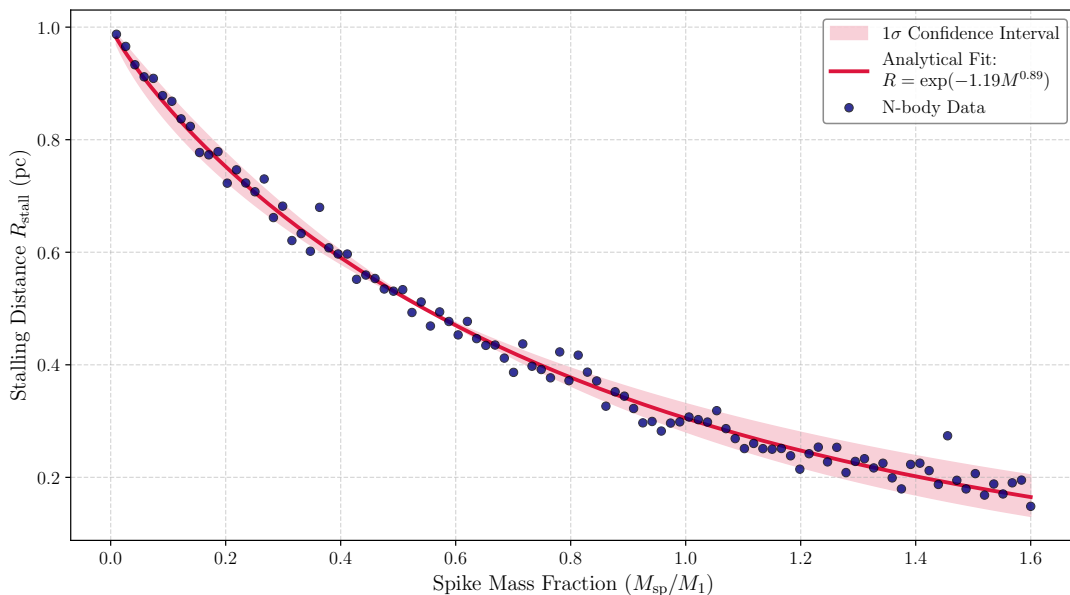


Figure 9: Stalling distance R_{stall} as a function of the total dark matter spike mass, expressed as the mass fraction M_{spike}/M_1 . The data points correspond to the binary separation at $t = 20000$ yr in the simulations shown in Fig. 7, which is used as a proxy for the final stalling distance. The red curve shows a stretched-exponential fit, $R_{\text{stall}} = \exp[-1.19(M_{\text{spike}}/M_1)^{0.89}]$, and the shaded region indicates the 1σ confidence interval. The agreement between the numerical data and the fit supports the interpretation that increasing the spike mass provides a larger reservoir of particles capable of extracting orbital energy from the binary before depletion occurs.

reservoir, but instead has a non-trivial radial density profile, the numerical data are better described by a stretched exponential of the form

$$R_{\text{stall}} = R_{\text{initial}} \exp \left[-C \left(\frac{M_{\text{spike}}}{M_1} \right)^\alpha \right].$$

The fit shown in Fig. 9 gives $\alpha \simeq 0.89$, close to the linear exponential expectation.

The monotonic decrease of R_{stall} confirms the expected reservoir effect: a more massive spike contains more particles that can be ejected through slingshot interactions, allowing the binary to lose more orbital energy before the local dark matter distribution is depleted. However, even in this idealized collisionless setup, the existence of a finite stalling radius shows that increasing the initial CDM reservoir does not remove the backreaction problem; it only delays it. This motivates the SIDM scenario, where self-interactions can continuously replenish the central region instead of relying on a finite, exhaustible spike.

6 Conclusions

In summary, the Final Parsec Problem arises because the known hardening mechanisms operate efficiently in different separation regimes but do not, by themselves, robustly bridge the gap between the stellar-dynamical and gravitational-wave-dominated phases. At large separations, stellar scattering and environmental torques can extract orbital energy from the supermassive black hole binary. However, in approximately spherical

collisionless systems, stellar hardening eventually becomes inefficient due to loss-cone depletion. At very small separations, gravitational-wave emission rapidly drives the final coalescence, but it is too weak to dominate at parsec-scale separations. Standard CDM dynamical friction can initially provide an efficient intermediate mechanism, but the binary injects enough orbital energy into the dark matter spike to heat, expand, and ultimately deplete it. Thus, a collisionless CDM spike does not provide a self-sustaining solution to the Final Parsec Problem.

The numerical toy model developed in this work illustrates this collisionless depletion mechanism explicitly. In the discrete-particle CDM limit, the binary hardens through individual gravitational slingshot encounters: dark matter particles gain kinetic energy and are ejected from the central region, while the binary loses orbital energy and becomes more tightly bound. Coarse-grained over many particles, these microscopic slingshots appear as an effective dynamical-friction force acting on the black holes. However, the same process that produces the drag also removes the particles that generate it. Once the local reservoir has been heated or ejected, the effective dynamical friction weakens and the binary stalls. This provides a direct microscopic interpretation of the backreaction problem of CDM spikes.

Self-Interacting Dark Matter offers a natural way to avoid this depletion. In the SIDM scenario, particle self-interactions thermalize the inner halo and create a central core that acts as a large heat and particle reservoir. As the binary heats the central spike, SIDM collisions transport energy and repopulate the inner region, allowing the effective dynamical friction to persist instead of shutting off. In particular, the massive-mediator SIDM model provides a viable balance: it supplies enough central density to drive a sub-Gyr inspiral, while maintaining a sufficiently bound and replenished reservoir to survive the binary’s energy injection.

A key observational consequence of this mechanism is a low-frequency turnover, or softening, in the stochastic gravitational-wave background. At large separations, a substantial fraction of the binary’s orbital energy is dissipated into the dark matter environment rather than emitted as gravitational waves, suppressing the gravitational-wave spectrum at low frequencies. This feature is consistent with current Pulsar Timing Array measurements and provides a testable signature of environmental hardening. Future PTA data, together with complementary gravitational-wave observations and improved modeling of supermassive black hole binary environments, will be crucial for constraining the allowed SIDM parameter space.

Moving forward, the next theoretical step is to replace the collisionless toy model used here with high-resolution simulations that include a genuinely collisional SIDM component. Such simulations should test whether self-interactions can dynamically replenish the spike while the binary hardens, and whether the system can consistently bridge the final parsec without exhausting the dark matter reservoir.

A Derivation of the Isothermal Poisson Equation

The isothermal sphere is a fundamental model in astrophysics, often used to describe the steady-state distribution of collisionless systems like dark matter halos or stellar clusters where the velocity dispersion is constant throughout the system.

The derivation requires three fundamental physical principles: spherical symmetry, hydrostatic equilibrium, and Poisson's equation for gravity.

A.1 Hydrostatic Equilibrium

For a spherically symmetric, self-gravitating system in steady-state equilibrium, the inward pull of gravity is perfectly balanced by the outward "pressure" of the particles (driven by their velocity dispersion). This is expressed by the hydrostatic equilibrium equation (or the spherical Jeans equation for a non-rotating, isotropic system):

$$\frac{dP}{dr} = -\rho \frac{d\Phi}{dr} \quad (25)$$

where P is the effective pressure, ρ is the mass density, and Φ is the gravitational potential.

A.2 The Isothermal Equation of State

For an isothermal system, the 1D velocity dispersion σ is constant with radius. We can define an effective equation of state analogous to an ideal gas:

$$P = \rho \sigma^2 \quad (26)$$

Substituting Equation 26 into Equation 25, we obtain:

$$\sigma^2 \frac{d\rho}{dr} = -\rho \frac{d\Phi}{dr} \quad (27)$$

Dividing by ρ yields a separable differential equation:

$$\sigma^2 \frac{d\rho}{\rho} = -d\Phi \quad (28)$$

Integrating both sides from the center of the system ($r = 0$, where $\rho = \rho_c$ and $\Phi = \Phi_c$) to an arbitrary radius r :

$$\sigma^2 \ln \left(\frac{\rho}{\rho_c} \right) = -(\Phi - \Phi_c) \quad (29)$$

Solving for the density profile $\rho(r)$, we recover the standard Maxwell-Boltzmann distribution for an isothermal potential:

$$\rho(r) = \rho_c \exp \left(-\frac{\Phi - \Phi_c}{\sigma^2} \right) \quad (30)$$

A.3 Poisson's Equation

The gravitational potential is related to the density via the spherical Poisson equation:

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 4\pi G \rho \quad (31)$$

To combine the thermodynamics with the gravity, we define a dimensionless relative potential, ψ :

$$\psi \equiv \frac{\Phi - \Phi_c}{\sigma^2} \implies \Phi = \Phi_c + \sigma^2 \psi \quad (32)$$

Note that at the center ($r = 0$), $\psi = 0$. By definition, Equation 30 simplifies to:

$$\rho = \rho_c e^{-\psi} \quad (33)$$

Substituting Φ into Poisson's equation (Equation 31), we get:

$$\frac{\sigma^2}{r^2} \frac{d}{dr} \left(r^2 \frac{d\psi}{dr} \right) = 4\pi G \rho_c e^{-\psi} \quad (34)$$

A.4 Non-Dimensionalization

To strip the equation of its physical units, we introduce a characteristic length scale known as the core radius (or King radius), r_c :

$$r_c \equiv \sqrt{\frac{\sigma^2}{4\pi G \rho_c}} \quad (35)$$

We then define a dimensionless radial coordinate ξ :

$$\xi \equiv \frac{r}{r_c} \implies r = r_c \xi \quad (36)$$

Substituting r with $r_c \xi$ transforms the differential operator:

$$\frac{\sigma^2}{r_c^2 \xi^2} \frac{1}{r_c} \frac{d}{d\xi} \left(r_c^2 \xi^2 \frac{1}{r_c} \frac{d\psi}{d\xi} \right) = 4\pi G \rho_c e^{-\psi} \quad (37)$$

The r_c terms inside the derivative cancel out. Factoring out the remaining constants yields:

$$\left(\frac{\sigma^2}{r_c^2} \right) \frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\psi}{d\xi} \right) = 4\pi G \rho_c e^{-\psi} \quad (38)$$

By the definition of r_c , the term (σ^2/r_c^2) is exactly equal to $4\pi G \rho_c$. Therefore, the constants on both sides cancel completely, leaving the elegant, dimensionless form:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\psi}{d\xi} \right) = e^{-\psi} \quad (39)$$

This is the standard Isothermal Poisson Equation. To solve it numerically, one must provide the central boundary conditions: $\psi(0) = 0$ (by definition) and $\psi'(0) = 0$ (to ensure the density gradient is flat at the origin).

B Dynamical Friction: A Fluid Dynamics Approach

Let us consider an infinite, uniform sea of Cold Dark Matter (CDM) with a constant background density ρ_0 . A massive particle of mass M moves through this medium with a constant velocity $\vec{v}$. We aim to determine the dynamical friction force exerted on this mass by the dark matter wake it induces.

The total mass density of the system is the sum of the dark matter density and the external mass:

$$\rho_{\text{total}}(\vec{r}, t) = \rho_{\text{CDM}}(\vec{r}, t) + \rho_{\text{ext}}(\vec{r}, t) \quad (40)$$

The dark matter density can be expressed as a small perturbation $\delta\rho$ over the background density, such that $\rho_{\text{CDM}}(\vec{r}, t) = \rho_0 + \delta\rho(\vec{r}, t)$. The external mass is modeled as a moving point mass using a Dirac delta function: $\rho_{\text{ext}}(\vec{r}, t) = M\delta(\vec{r} - \vec{v}t)$.

The system is governed by the linearized fluid equations. First, we state the Poisson equation for the gravitational potential Φ :

$$\nabla^2\Phi = 4\pi G(\rho_0 + \delta\rho + M\delta(\vec{r} - \vec{v}t)) \quad (41)$$

Next, the continuity equation for the dark matter fluid. Assuming the induced velocity $\vec{u}$ in the fluid is small, we linearize it as follows:

$$\frac{\partial\delta\rho}{\partial t} + \nabla \cdot (\rho_0\vec{u}) = 0 \implies \nabla \cdot \vec{u} = -\frac{1}{\rho_0} \frac{\partial\delta\rho}{\partial t} \quad (42)$$

Finally, we use the Euler equation for a pressureless fluid (representing cold dark matter). Because the perturbations are small, we can neglect the non-linear convective term $(\vec{u} \cdot \nabla)\vec{u} \approx 0$:

$$\frac{\partial\vec{u}}{\partial t} = -\nabla\Phi \quad (43)$$

To obtain a single differential equation for the density perturbation, we take the divergence of the linearized Euler equation:

$$\frac{\partial}{\partial t}(\nabla \cdot \vec{u}) = -\nabla^2\Phi \quad (44)$$

Substituting the divergence of the velocity from the continuity equation, we find:

$$\frac{\partial}{\partial t} \left(-\frac{1}{\rho_0} \frac{\partial\delta\rho}{\partial t} \right) = -\nabla^2\Phi \implies \frac{1}{\rho_0} \frac{\partial^2\delta\rho}{\partial t^2} = \nabla^2\Phi \quad (45)$$

Inserting the Poisson equation into this result yields:

$$\frac{\partial^2\delta\rho}{\partial t^2} = 4\pi G\rho_0(\rho_0 + \delta\rho + M\delta(\vec{r} - \vec{v}t)) \quad (46)$$

At this stage, we apply the "Jeans swindle". We ignore the uniform background gravitational field generated by ρ_0 , as an infinite, homogeneous universe does not undergo global collapse in this idealized Newtonian framework. The term $4\pi G\rho_0^2$ cancels out, reducing the expression to a driven wave equation for the density perturbation:

$$\frac{\partial^2\delta\rho}{\partial t^2} - 4\pi G\rho_0\delta\rho = 4\pi G\rho_0 M\delta(\vec{r} - \vec{v}t) \quad (47)$$

B.1 Solution in Fourier Space

With the differential equation for the density perturbation $\delta\rho$ established, we can solve it to find the density increase in the wake, and subsequently the drag force exerted on M . To solve for $\delta\rho$, it is highly advantageous to transform the equation into Fourier space.

We define the Fourier transform of the density perturbation as:

$$\mathcal{F}[\delta\rho(\vec{r}, t)] = \frac{1}{(2\pi)^{3/2}} \int \delta\rho(\vec{r}, t) e^{i\vec{k}\cdot\vec{r}} d^3r \equiv \tilde{\delta\rho}(\vec{k}, t) \quad (48)$$

Applying this transform to our driven wave equation, the spatial Dirac delta function is integrated out:

$$\frac{\partial^2 \tilde{\delta\rho}}{\partial t^2} - 4\pi G\rho_0 \tilde{\delta\rho} = \frac{4\pi GM\rho_0}{(2\pi)^{3/2}} \int \delta(\vec{r} - \vec{v}t) e^{i\vec{k}\cdot\vec{r}} d^3r \quad (49)$$

Evaluating the integral yields $e^{i\vec{k}\cdot\vec{v}t}$. We can group the constants on the right-hand side into a single term C :

$$C \equiv \frac{4\pi GM\rho_0}{(2\pi)^{3/2}} \quad (50)$$

And we can define another constant C' for the background self-gravity term (related to the Jeans frequency):

$$C' \equiv 4\pi G\rho_0 \quad (51)$$

Substituting these constants, the equation simplifies to a driven harmonic oscillator in Fourier space:

$$\frac{\partial^2 \tilde{\delta\rho}}{\partial t^2} - C' \tilde{\delta\rho} = C e^{i\vec{k}\cdot\vec{v}t} \quad (52)$$

To solve this non-homogeneous ordinary differential equation, we propose an exponential *ansatz* for the particular solution:

$$\tilde{\delta\rho} = A e^{i\vec{k}\cdot\vec{v}t} \quad (53)$$

Taking the second time derivative of the ansatz and substituting it back into the differential equation gives:

$$A [(i\vec{k} \cdot \vec{v})^2 - C'] e^{i\vec{k}\cdot\vec{v}t} = C e^{i\vec{k}\cdot\vec{v}t} \quad (54)$$

Since $(i\vec{k} \cdot \vec{v})^2 = -(\vec{k} \cdot \vec{v})^2$, we can cancel the exponentials and solve for the amplitude A :

$$A [-(\vec{k} \cdot \vec{v})^2 - C'] = C \implies A = \frac{-C}{(\vec{k} \cdot \vec{v})^2 + C'} \quad (55)$$

Thus, the particular solution for the density perturbation in Fourier space is:

$$\tilde{\delta\rho}(\vec{k}, t) = \frac{-C}{(\vec{k} \cdot \vec{v})^2 + C'} e^{i\vec{k}\cdot\vec{v}t} \quad (56)$$

B.2 Inverse Fourier Transform and Complex Integration

To find the density perturbation in real space, we apply the inverse Fourier transform:

$$\delta\rho(\vec{r}, t) = \frac{1}{(2\pi)^{3/2}} \int \tilde{\delta\rho}(\vec{k}, t) e^{-i\vec{k}\cdot\vec{r}} d^3k \quad (57)$$

Substituting our particular solution $\tilde{\delta\rho}$:

$$\delta\rho(\vec{r}, t) = -\frac{C}{(2\pi)^{3/2}} \int \frac{e^{i\vec{k}\cdot\vec{v}t} e^{-i\vec{k}\cdot\vec{r}}}{(\vec{k}\cdot\vec{v})^2 + C'} d^3k = -\frac{C}{(2\pi)^{3/2}} \int \frac{e^{-i\vec{k}\cdot(\vec{r}-\vec{v}t)}}{(\vec{k}\cdot\vec{v})^2 + C'} d^3k \quad (58)$$

To solve this 3D integral, it is highly convenient to align our coordinate system with the direction of motion. Let the velocity of the mass M be along the z-axis: $\vec{v} = (0, 0, v)$. Consequently, $\vec{k}\cdot\vec{v} = k_z v$.

We define the relative position vector $\vec{r}' = \vec{r} - \vec{v}t$. In cylindrical coordinates, this is $\vec{r}' = (r_\perp \cos \theta', r_\perp \sin \theta', z')$. We also express the wave vector $\vec{k}$ in cylindrical coordinates: $d^3k = k_\perp dk_\perp d\phi_k dk_z$.

The integral can now be separated into a parallel component (z) and a perpendicular component ($\perp$):

$$\delta\rho(\vec{r}') = -\frac{C}{(2\pi)^{3/2}} \int_0^\infty k_\perp dk_\perp \int_0^{2\pi} e^{-i\vec{k}_\perp\cdot\vec{r}'_\perp} d\phi_k \int_{-\infty}^\infty \frac{e^{-ik_z z'}}{k_z^2 v^2 + C'} dk_z \quad (59)$$

Let us focus on the integral over k_z , which we will denote as I_z :

$$I_z = \int_{-\infty}^\infty \frac{e^{-ik_z z'}}{v^2 (k_z^2 + \frac{C'}{v^2})} dk_z \quad (60)$$

This integral can be solved using Cauchy's Residue Theorem by extending k_z into the complex plane. The integrand has two simple poles located at:

$$k_z = \pm i \frac{\sqrt{C'}}{v} \quad (61)$$

We are interested in the dark matter wake, which forms *behind* the moving mass ($z' < 0$). According to Jordan's Lemma, to ensure that the exponential term $e^{-ik_z z'}$ decays to zero along the infinite arc, we must close our contour in the **upper half-plane** (since z' is negative, a positive imaginary part of k_z yields a decaying real exponential).

By closing the contour in the upper half-plane (traversed counter-clockwise), we enclose only the pole at $z_0 = +i \frac{\sqrt{C'}}{v}$.

B.3 The Unphysical Void and Jeans Instability

Using the Residue Theorem, we evaluate I_z at the enclosed pole $k_z = +i \frac{\sqrt{C'}}{v}$:

$$I_z = 2\pi i \lim_{k_z \rightarrow i \frac{\sqrt{C'}}{v}} \left[\frac{e^{-ik_z z'}}{v^2 (k_z + i \frac{\sqrt{C'}}{v})} \right] = 2\pi i \frac{e^{\frac{\sqrt{C'}}{v} z'}}{v^2 (2i \frac{\sqrt{C'}}{v})} = \frac{\pi e^{\frac{\sqrt{C'}}{v} z'}}{v \sqrt{C'}} \quad (62)$$

Next, we solve the perpendicular integrals over $k_\perp$ and ϕ_k . The angular integral results in a Bessel function of the first kind of order zero, J_0 :

$$I' = \int_0^\infty k_\perp dk_\perp \int_0^{2\pi} \exp[-ik_\perp r'_\perp \cos(\phi_k - \theta')] d\phi_k = 2\pi \int_0^\infty k_\perp J_0(k_\perp r'_\perp) dk_\perp \quad (63)$$

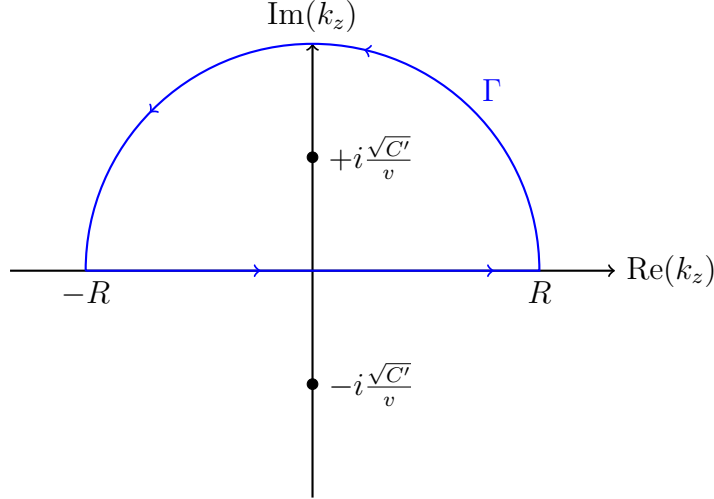


Figure 10: Complex integration contour for $z' < 0$. The contour is closed in the upper half-plane, enclosing the positive imaginary pole.

Combining all parts, the density perturbation behind the mass is:

$$\delta\rho(\vec{r}') = -\frac{C}{(2\pi)^{3/2}} \frac{\pi e^{\frac{\sqrt{C'}}{v} z'}}{v\sqrt{C'}} 2\pi \int_0^\infty k_\perp J_0(k_\perp r'_\perp) dk_\perp \quad (64)$$

Notice the overall **negative sign**. Because the constants and the integrals yield positive values, this result dictates that $\delta\rho < 0$. Physically, this means a "void" or underdensity forms behind the mass, which contradicts the attractive nature of gravity.

This unphysical result arises because we included the self-interaction of the dark matter fluid via the C' term. In a perfectly cold, pressureless fluid, self-gravity causes a global collapse known as the **Jeans instability**, making the stationary wake solution invalid.

B.4 Removing Self-Interaction

To properly model the dynamical friction wake, we must assume that the background dark matter particles are solely influenced by the external mass M , and do not interact gravitationally with each other. Thus, we modify the Poisson equation by neglecting the self-gravity of the perturbation:

$$\nabla^2\Phi = 4\pi G\rho_{\text{ext}} = 4\pi GM\delta(\vec{r} - \vec{v}t) \quad (65)$$

Substituting this into our fluid equations (effectively setting $C' = 0$), the wave equation becomes:

$$\frac{\partial^2 \delta\rho}{\partial t^2} = 4\pi G\rho_0 M\delta(\vec{r} - \vec{v}t) \quad (66)$$

Applying the Fourier transform as before, we obtain:

$$\frac{\partial^2 \tilde{\delta\rho}}{\partial t^2} = C e^{i\vec{k}\cdot\vec{v}t} \quad (67)$$

We use the same exponential ansatz, $\tilde{\delta\rho} = A e^{i\vec{k}\cdot\vec{v}t}$, which yields:

$$-(\vec{k} \cdot \vec{v})^2 A = C \implies A = -\frac{C}{(\vec{k} \cdot \vec{v})^2} \quad (68)$$

Therefore, our new particular solution in Fourier space is:

$$\tilde{\delta\rho}(\vec{k}, t) = -\frac{C}{(\vec{k} \cdot \vec{v})^2} e^{i\vec{k} \cdot \vec{v}t} \quad (69)$$

B.5 Return to Real Space and Causality Prescription

With the self-gravity of the background removed, our density perturbation in Fourier space is $\tilde{\delta\rho}(\vec{k}, t) = -\frac{C}{(\vec{k} \cdot \vec{v})^2} e^{i\vec{k} \cdot \vec{v}t}$. We apply the inverse Fourier transform to return to real space:

$$\delta\rho(\vec{r}, t) = -\frac{C}{(2\pi)^{3/2}} \int \frac{e^{i\vec{k} \cdot \vec{v}t} e^{-i\vec{k} \cdot \vec{r}}}{(\vec{k} \cdot \vec{v})^2} d^3k = -\frac{C}{(2\pi)^{3/2}} \int \frac{e^{-i\vec{k} \cdot \vec{r}'}}{(\vec{k} \cdot \vec{v})^2} d^3k \quad (70)$$

where $\vec{r}' = \vec{r} - \vec{v}t$. Aligning the velocity vector such that $\vec{v} = (0, 0, v)$ and switching to cylindrical coordinates in k -space, we separate the integral:

$$\delta\rho(\vec{r}') = -\frac{C}{(2\pi)^{3/2}} \int_0^\infty k_\perp dk_\perp \int_0^{2\pi} e^{-i\vec{k}_\perp \cdot \vec{r}'_\perp} d\phi_k \int_{-\infty}^\infty \frac{e^{-ik_z z'}}{(k_z v)^2} dk_z \quad (71)$$

The integral over k_z now features a double pole exactly at the origin ($k_z = 0$). To resolve this and enforce **causality** (ensuring the wake forms only *behind* the mass), we introduce the $i\epsilon$ prescription. We give the frequency a small, positive imaginary part by replacing k_z with $(k_z - i\epsilon)$. This shifts the pole slightly into the upper half-plane:

$$I_z = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^\infty \frac{e^{-ik_z z'}}{v^2(k_z - i\epsilon)^2} dk_z \quad (72)$$

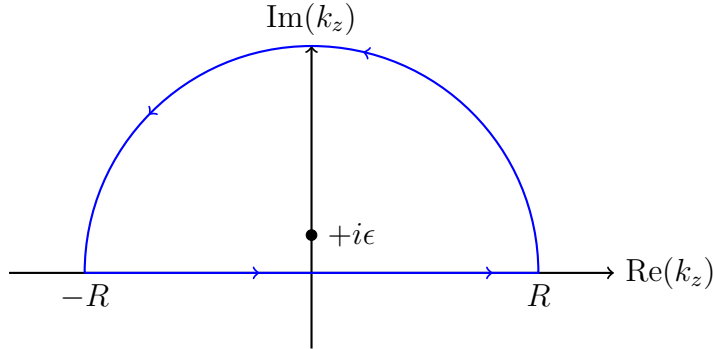


Figure 11: Contour integration for the k_z integral using the $i\epsilon$ causality prescription. The double pole is shifted to the upper half-plane. Since we evaluate the wake ($z' < 0$), we close the contour in the upper half-plane to ensure exponential decay.

Because we are evaluating the region behind the mass ($z' < 0$), we close the contour in the upper half-plane to satisfy Jordan's Lemma. The contour encloses the pole of order $m = 2$ at $z_0 = i\epsilon$. We calculate its residue:

$$\text{Res}(z_0) = \frac{1}{(2-1)!} \lim_{k_z \rightarrow i\epsilon} \frac{d}{dk_z} \left[(k_z - i\epsilon)^2 \frac{e^{-ik_z z'}}{v^2(k_z - i\epsilon)^2} \right] \quad (73)$$

$$\text{Res}(z_0) = \lim_{k_z \rightarrow i\epsilon} \frac{d}{dk_z} \left(\frac{e^{-ik_z z'}}{v^2} \right) = \lim_{k_z \rightarrow i\epsilon} \left(-\frac{iz'}{v^2} e^{-ik_z z'} \right) = -\frac{iz'}{v^2} e^{\epsilon z'} \quad (74)$$

Applying the Residue Theorem for the counter-clockwise contour:

$$I_z = 2\pi i \text{Res}(z_0) = 2\pi i \left(-\frac{iz'}{v^2} e^{\epsilon z'} \right) = \frac{2\pi z'}{v^2} e^{\epsilon z'} \quad (75)$$

Taking the limit as $\epsilon \rightarrow 0$, we find $I_z = \frac{2\pi z'}{v^2}$.

Substituting I_z back into the density perturbation equation:

$$\delta\rho(\vec{r}') = -\frac{C}{(2\pi)^{3/2}} \frac{2\pi z'}{v^2} \int_0^\infty k_\perp dk_\perp \int_0^{2\pi} e^{-i\vec{k}_\perp \cdot \vec{r}'_\perp} d\phi_k \quad (76)$$

The angular integral evaluates to $2\pi J_0(k_\perp r_\perp)$. Assembling all the components, we arrive at the final real-space density perturbation:

$$\delta\rho(r_\perp, z') = -\frac{C}{(2\pi)^{3/2}} \frac{2\pi z'}{v^2} 2\pi \int_0^\infty k_\perp J_0(k_\perp r_\perp) dk_\perp \quad (77)$$

$$\delta\rho(r_\perp, z') = -\sqrt{2\pi} C \frac{z'}{v^2} \int_0^\infty k_\perp J_0(k_\perp r_\perp) dk_\perp \quad (78)$$

B.6 The Ultraviolet Cutoff and Wake Density

Evaluating the integral over $k_\perp$ from 0 to ∞ yields a Dirac delta function:

$$\int_0^\infty k_\perp J_0(k_\perp r_\perp) dk_\perp = \frac{\delta(r_\perp)}{r_\perp} \quad (79)$$

This indicates that the density diverges to infinity exactly on the axis behind the mass ($r_\perp = 0$). Physically, an infinite density breaks our initial linearizing assumption that perturbations are small ($\delta\rho \ll \rho_0$).

Since our theory is linear, we cannot allow background particles to approach arbitrarily close to the massive particle M . At very close distances, the gravitational potential energy dominates over the kinetic energy, and the trajectories bend violently. We estimate this minimum impact parameter, r_{min} , by equating the kinetic energy to the gravitational potential energy:

$$\frac{1}{2}mv^2 \approx \frac{GMm}{r_{min}} \implies r_{min} = \frac{2GM}{v^2} \quad (80)$$

To cure the divergence, we introduce an ultraviolet cutoff in Fourier space, k_{max} , corresponding to this minimum distance:

$$k_{max} = \frac{1}{r_{min}} \quad (81)$$

We now evaluate the density integral up to this cutoff:

$$\int_0^{k_{max}} k_\perp J_0(k_\perp r_\perp) dk_\perp \quad (82)$$

We apply the substitution $x = k_\perp r_\perp$, which gives $dx = r_\perp dk_\perp$ and changes the upper limit to $x_{max} = k_{max} r_\perp$. The integral becomes:

$$\frac{1}{r_\perp^2} \int_0^{k_{max} r_\perp} x J_0(x) dx \quad (83)$$

Using the fundamental identity for Bessel functions, $\frac{d}{dx}[xJ_1(x)] = xJ_0(x)$, we can easily integrate:

$$\frac{1}{r_\perp^2} \left[xJ_1(x) \right]_0^{k_{max}r_\perp} = \frac{k_{max}r_\perp}{r_\perp^2} J_1(k_{max}r_\perp) = \frac{k_{max}}{r_\perp} J_1(k_{max}r_\perp) \quad (84)$$

Substituting this analytical result back into our equation for the density perturbation, we find the complete density field of the dark matter wake:

$$\delta\rho(r_\perp, z') = -\sqrt{2\pi}C \frac{z'}{v^2} \frac{k_{max}}{r_\perp} J_1(k_{max}r_\perp) \quad (85)$$

B.7 The Dynamical Friction Force

With the density of the wake known, we can calculate the total gravitational force it exerts back on the moving mass M . From Newton's law of gravitation, the force is the integral of the density over the entire volume. However, only the component of the force along the axis of motion (z') contributes to the drag.

From geometry, the projection angle α provides the necessary component:

$$\cos \alpha = \frac{z'}{|\vec{r}'|} = \frac{z'}{\sqrt{r_\perp^2 + z'^2}} \quad (86)$$

The dynamical friction force F_{DF} along the z -axis is:

$$F_{DF} = \int \frac{GM\delta\rho(\vec{r}')}{|\vec{r}'|^2} \cos \alpha d^3r' = \int \frac{GM\delta\rho(\vec{r}')z'}{|\vec{r}'|^3} d^3r' \quad (87)$$

Substituting our expression for $\delta\rho$ (keeping its intrinsic negative sign) and the spherical volume element in cylindrical coordinates ($d^3r' = r_\perp dr_\perp dz' d\theta$), we get a strictly negative prefactor:

$$F_{DF} = -\frac{\sqrt{2\pi}C}{v^2} GMk_{max} \int \frac{z'^2 J_1(k_{max}r_\perp)}{r_\perp |\vec{r}'|^3} d^3r' \quad (88)$$

Substituting the volume element and performing the angular integration ($\int d\theta = 2\pi$), the $r_\perp$ terms partially cancel out. Also, to avoid a divergence from an infinitely massive wake, we must integrate z' over a finite trail length, from $-\bar{z}$ (where $\bar{z} = vt$, the distance traveled) up to 0:

$$F_{DF} = -2\pi \frac{\sqrt{2\pi}C}{v^2} GMk_{max} \int_0^\infty J_1(k_{max}r_\perp) dr_\perp \int_{-\bar{z}}^0 \frac{z'^2}{(r_\perp^2 + z'^2)^{3/2}} dz' \quad (89)$$

Let us solve the z' integral using trigonometric substitution. We set $z' = r_\perp \tan \theta$, which gives $dz' = \frac{r_\perp}{\cos^2 \theta} d\theta$. The denominator transforms as $(r_\perp^2 + r_\perp^2 \tan^2 \theta)^{3/2} = \frac{r_\perp^3}{\cos^3 \theta}$. Substituting these into the integral:

$$\int \frac{(r_\perp \tan \theta)^2}{r_\perp^3 / \cos^3 \theta} \frac{r_\perp}{\cos^2 \theta} d\theta = \int \frac{\tan^2 \theta \cos^3 \theta}{\cos^2 \theta} d\theta = \int \frac{\sin^2 \theta}{\cos \theta} d\theta \quad (90)$$

Using the fundamental identity $\sin^2 \theta = 1 - \cos^2 \theta$, we can split the integral into standard forms:

$$\int \left(\frac{1}{\cos \theta} - \cos \theta \right) d\theta = \left[\ln |\sec \theta + \tan \theta| - \sin \theta \right]_{\bar{\theta}}^0 \quad (91)$$

Evaluating at the upper limit ($z' = 0 \implies \theta = 0$) yields exactly 0. For the lower limit ($z' = -\bar{z}$), we construct a right triangle where the opposite side is $-\bar{z}$, the adjacent is $r_\perp$, and the hypotenuse is $\sqrt{r_\perp^2 + \bar{z}^2}$. Therefore, $\sec \bar{\theta} = \frac{\sqrt{r_\perp^2 + \bar{z}^2}}{r_\perp}$ and $\sin \bar{\theta} = \frac{-\bar{z}}{\sqrt{r_\perp^2 + \bar{z}^2}}$.

Subtracting the lower limit evaluation from zero:

$$- \left[\ln \left| \frac{\sqrt{r_\perp^2 + \bar{z}^2}}{r_\perp} - \frac{\bar{z}}{r_\perp} \right| - \left(\frac{-\bar{z}}{\sqrt{r_\perp^2 + \bar{z}^2}} \right) \right] \quad (92)$$

We rationalize the argument of the logarithm by multiplying the numerator and denominator by the conjugate ($\sqrt{r_\perp^2 + \bar{z}^2} + \bar{z}$):

$$\ln \left[\frac{\sqrt{r_\perp^2 + \bar{z}^2} + \bar{z}}{r_\perp} \right] - \frac{\bar{z}}{\sqrt{r_\perp^2 + \bar{z}^2}} \quad (93)$$

Finally, we apply the astrophysical limit. The length of the wake is macroscopic and vastly larger than the transverse impact parameter ($\bar{z} \gg r_\perp$). Under this regime, $\sqrt{r_\perp^2 + \bar{z}^2} \approx \bar{z}$. Applying this approximation, the result simplifies beautifully to:

$$\approx \ln \left(\frac{2\bar{z}}{r_\perp} \right) - 1 \quad (94)$$

Substituting this approximation back into our radial integral, we need to evaluate:

$$2\pi \int_0^\infty J_1(k_{max} r_\perp) \left[\ln \left(\frac{2\bar{z}}{r_\perp} \right) - 1 \right] dr_\perp \quad (95)$$

To solve this rigorously and avoid diverging boundary terms from integration by parts, we use the substitution $x = k_{max} r_\perp$, giving $dr_\perp = dx/k_{max}$. Using logarithm properties $\ln(A/x) = \ln A - \ln x$:

$$\frac{2\pi}{k_{max}} \left[(\ln(2\bar{z}k_{max}) - 1) \int_0^\infty J_1(x) dx - \int_0^\infty J_1(x) \ln x dx \right] \quad (96)$$

We know that $\int_0^\infty J_1(x) dx = 1$. The second integral evaluates to a finite constant $\int_0^\infty J_1(x) \ln x dx = \ln 2 - \gamma$ (where γ is the Euler-Mascheroni constant). The exact analytical evaluation is:

$$\frac{2\pi}{k_{max}} \left[\ln(2\bar{z}k_{max}) - 1 - \ln 2 + \gamma \right] \approx \frac{2\pi}{k_{max}} \ln(\bar{z}k_{max}) \quad (97)$$

Since $\bar{z}$ is our macroscopic maximum wake length (r_{max}) and $k_{max} = 1/r_{min}$, this result is strictly positive and reduces to the universally renowned **Coulomb Logarithm**, denoted as $\ln \Lambda$:

$$\frac{2\pi}{k_{max}} \ln \left(\frac{r_{max}}{r_{min}} \right) \equiv \frac{2\pi}{k_{max}} \ln \Lambda \quad (98)$$

B.8 Final Conclusion: The Chandrasekhar Formula

We substitute this positive radial integral back into our force equation (which correctly holds the negative prefactor):

$$F_{DF} = -\frac{\sqrt{2\pi}C}{v^2} GM k_{max} \left(\frac{2\pi}{k_{max}} \ln \Lambda \right) = -\frac{(2\pi)^{3/2}C}{v^2} GM \ln \Lambda \quad (99)$$

Finally, we recall our initial definition for the constant $C = \frac{4\pi GM\rho_0}{(2\pi)^{3/2}}$. Substituting this back to restore the physical parameters:

$$F_{DF} = -\frac{(2\pi)^{3/2}}{v^2} \left(\frac{4\pi GM\rho_0}{(2\pi)^{3/2}} \right) GM \ln \Lambda \quad (100)$$

Simplifying the terms, we arrive at the exact analytical expression for the force:

$$F_{DF} = -\frac{4\pi G^2 M^2 \rho_0}{v^2} \ln \Lambda \quad (101)$$

This is the celebrated Chandrasekhar Dynamical Friction formula. Starting from the fundamental fluid dynamics equations of a pressureless medium, we have successfully derived the exact gravitational drag experienced by a massive body moving through a sea of dark matter.

C Dynamical Friction Power Loss for a Binary System

While Appendix B derives the dynamical friction force exerted on a single massive body, the Supermassive Black Hole (SMBH) system consists of a binary. Therefore, the total orbital power loss (P_{df}) must account for the individual drag forces acting on both the primary (M_1) and secondary (M_2) black holes as they orbit their common center of mass.

C.1 D.1 Binary Kinematics

Consider a circular binary with a total mass $M_\bullet = M_1 + M_2$ and a mass ratio defined as $q \equiv M_2/M_1 \leq 1$. The total orbital separation is $R = r_1 + r_2$, where r_1 and r_2 are the distances from each black hole to the center of mass. By definition:

$$r_1 = R \left(\frac{q}{1+q} \right), \quad r_2 = R \left(\frac{1}{1+q} \right) \quad (102)$$

The orbital velocities v_1 and v_2 are determined by equating the mutual gravitational force to the centripetal force for each body. For the secondary mass M_2 :

$$\frac{GM_1 M_2}{R^2} = M_2 \frac{v_2^2}{r_2} \implies v_2 = \sqrt{\frac{GM_1}{R(1+q)}} \quad (103)$$

Similarly, evaluating the primary mass M_1 yields $v_1 = qv_2$. Notice that the less massive secondary moves faster and occupies a wider orbit.

C.2 D.2 Total Power Dissipation

The power dissipated by dynamical friction for a single body is the dot product of the drag force and its velocity: $P_i = F_{df,i} \cdot v_i$. Using the standard Chandrasekhar formula (derived in Appendix B), the power loss for mass i is:

$$P_i = 4\pi G^2 M_i^2 \rho_{sp}(r_i) \ln \Lambda \left(\frac{N_i(< v_i)}{v_i} \right) \quad (104)$$

where $\rho_{\text{sp}}(r_i)$ is the local dark matter density evaluated at the specific orbital radius of the black hole, and $N_i(< v_i)$ is the fraction of dark matter particles moving slower than the black hole.

The total power loss of the binary is the sum of the dissipation from both components: $P_{\text{df}} = P_1 + P_2$. Substituting our kinematic relations ($M_2 = qM_1$ and $v_1 = qv_2$) into the power equations and factoring out the common terms yields the total power loss shown in Equation 12 of the main text:

$$P_{\text{df}} = 12\pi q^2 \sqrt{1+q} (GM_1)^{3/2} R^{1/2} \times \left[\frac{N_1(q)}{q^3} \rho_{\text{sp}}(r_1) + N_2(q) \rho_{\text{sp}}(r_2) \right] \quad (105)$$

(Note: The prefactor 12π absorbs the Coulomb logarithm $\ln \Lambda$ and geometric constants specific to the halo profile).

C.3 D.3 The Velocity Fraction $N(< v_{\text{bh}})$

The term $N(< v_{\text{bh}})$ physically represents the fraction of background particles that contribute to the drag. Particles moving much faster than the black hole do not effectively participate in forming the gravitational wake. Assuming the dark matter follows a Maxwell-Boltzmann velocity distribution $P(u)$ with a 1D velocity dispersion σ , the fraction is found by integrating over all speeds up to the black hole's velocity v_{bh} :

$$N(< v_{\text{bh}}) = \int_0^{v_{\text{bh}}} P(u) du = \sqrt{\frac{2}{\pi}} \frac{1}{\sigma^3} \int_0^{v_{\text{bh}}} u^2 e^{-u^2/2\sigma^2} du \quad (106)$$

Integrating this by parts yields a solution dependent on the error function (erf):

$$N(< v_{\text{bh}}) = \text{erf}\left(\frac{v_{\text{bh}}}{\sqrt{2}\sigma}\right) - \sqrt{\frac{2}{\pi}} \left(\frac{v_{\text{bh}}}{\sigma}\right) e^{-v_{\text{bh}}^2/2\sigma^2} \quad (107)$$

This analytical result explicitly dictates the drag efficiency differences between Cold Dark Matter (CDM) and Self-Interacting Dark Matter (SIDM).

- **For standard CDM:** The environment is kinematically cold ($\sigma \ll v_{\text{bh}}$). The argument of the error function approaches infinity, meaning $\text{erf}(\infty) \rightarrow 1$, and the exponential term vanishes. Thus, virtually all particles contribute to the drag ($N \approx 1$).
- **For SIDM:** The collisional nature of the particles thermalizes the core, raising the velocity dispersion σ so that it is comparable to, or strictly greater than, the binary's orbital velocity. Evaluating the function in this kinematically "hot" regime severely suppresses the fraction of participating particles, yielding $N \simeq 0.2$.

D Gravitational Wave Frequency and Energy Spectrum

To model the continuous gravitational wave spectrum emitted by the supermassive black hole binary, we must map the orbital dynamics to the emitted frequency and calculate the rate of energy dissipation.

D.1 1. Gravitational Wave Frequency

For a binary in a circular orbit, gravitational wave emission is dominated by the quadrupole moment. Consequently, the gravitational wave frequency (f_s) is exactly twice the orbital frequency ($f_{\text{orb}} = \omega/2\pi$). Using Kepler's third law for a total binary mass $M_{\bullet} = M_1 + M_2 = M_1(1 + q)$, the frequency is given by:

$$f_s = 2 \left(\frac{\omega}{2\pi} \right) = \frac{1}{\pi} \left(\frac{GM_{\bullet}}{R^3} \right)^{1/2} \quad (108)$$

To frame this in the context of the dark matter spike, we introduce a dimensionless separation parameter $x \equiv R/(2r_{\text{sp}})$, such that $R = 2r_{\text{sp}}x$. Substituting this into the frequency yields:

$$f_s = \frac{1}{\pi} \left(\frac{GM_{\bullet}}{(2r_{\text{sp}}x)^3} \right)^{1/2} = \frac{1}{2\pi} \left(\frac{GM_1(1 + q)}{2r_{\text{sp}}^3 x^3} \right)^{1/2} \quad (109)$$

We can further simplify this by defining the characteristic dynamical timescale of the spike as $t_{\text{sp}} \equiv (r_{\text{sp}}^3/GM_1)^{1/2}$. Substituting t_{sp} into the expression, we arrive at the final operational form for the gravitational wave frequency:

$$f_s = \frac{1}{2\pi t_{\text{sp}}} \sqrt{\frac{1 + q}{2x^3}} \quad (110)$$

D.2 2. Orbital Energy Spectrum

To derive the spectral shape of the emitted gravitational waves, we evaluate the rate of change of the orbital energy with respect to the frequency. The standard Newtonian orbital energy of the binary is:

$$E_{\text{orb}} = -\frac{GM_1 M_2}{2R} = -\frac{GM_1 M_2}{4x r_{\text{sp}}} \quad (111)$$

Taking the derivative of the orbital energy with respect to the dimensionless separation x yields:

$$\frac{dE_{\text{orb}}}{dx} = \frac{GM_1 M_2}{4x^2 r_{\text{sp}}} \quad (112)$$

From Equation 110, we know that $f_s \propto x^{-3/2}$. Taking the logarithmic derivative gives the relation between x and f_s :

$$\frac{dx}{df_s} = -\frac{2}{3} \frac{x}{f_s} \quad (113)$$

Applying the chain rule, we combine these terms to find the total orbital energy dissipated per unit frequency interval:

$$\frac{dE_{\text{orb}}}{df_s} = \frac{dE_{\text{orb}}}{dx} \frac{dx}{df_s} = \left(\frac{GM_1 M_2}{4x^2 r_{\text{sp}}} \right) \left(-\frac{2}{3} \frac{x}{f_s} \right) = -\frac{GM_1 M_2}{6f_s x r_{\text{sp}}} \quad (114)$$

Substituting $M_2 = qM_1$ and taking the absolute magnitude, the total spectral energy loss is:

$$\left| \frac{dE_{\text{orb}}}{df_s} \right| = \frac{qGM_1^2}{6f_s x r_{\text{sp}}} \quad (115)$$

3. Gravitational Wave Branching and Cutoff

The expression above represents the *total* energy drained from the binary. However, in a dark matter-rich environment, this energy loss is split between gravitational wave emission (P_{gw}) and dynamical friction (P_{df}). To isolate the spectrum of the gravitational waves, we apply a branching ratio representing the fraction of power radiated purely as waves:

$$\frac{dE_{\text{gw}}}{df_s} = \frac{qGM_1^2}{6f_s x r_{\text{sp}}} \left(\frac{P_{\text{gw}}}{P_{\text{gw}} + P_{\text{df}}} \right) \quad (116)$$

Finally, this spectrum is observationally truncated at the minimum physical separation, marking the plunge phase of the black holes. Adopting natural units where the speed of light is $c = 1$, this cutoff corresponds to the Schwarzschild radius of the combined black hole mass:

$$R_{\text{min}} = 2GM_{\bullet} = 2GM_1(1 + q) \quad (117)$$

E Analytical Derivation of the Exponential Stalling Law

While the numerical data clearly exhibits an exponential decay in the stalling radius as a function of the spike mass, we can rigorously justify this functional form by analyzing the differential energy exchange during the three-body slingshot encounters.

First, we consider the orbital energy of the supermassive black hole binary, which is inversely proportional to its separation, R :

$$E_{\text{bin}} \propto -\frac{1}{R} \quad (118)$$

If the binary shrinks by an infinitesimal amount dR , the corresponding change in its orbital binding energy is given by the derivative:

$$dE_{\text{bin}} \propto \frac{1}{R^2} dR \quad (119)$$

As the binary shrinks, it must eject a differential mass element of dark matter, dM , to conserve energy. In the slingshot mechanism, the kinetic energy of the ejected particles is proportional to the local Keplerian potential well of the binary. The ejection velocity scales with the orbital velocity, $v_{\text{ej}} \propto v_{\text{orb}} \propto R^{-1/2}$. Therefore, the kinetic energy carried away by the ejected mass element is:

$$dE_{\text{ej}} \propto dM \cdot v_{\text{ej}}^2 \propto \frac{dM}{R} \quad (120)$$

By conservation of energy, the energy lost by the binary's orbit must equal the kinetic energy gained by the ejected dark matter. To keep our formulation universally scalable, we normalize the ejected dark matter mass by the primary black hole mass, defining the mass fraction $\mu \equiv M_{\text{sp}}/M_1$. Equating the two energy differentials yields:

$$\frac{1}{R^2} dR = -C \frac{d\mu}{R} \quad (121)$$

where C is a dimensionless constant parameterizing the efficiency of the three-body scattering process, and the negative sign indicates that an ejection of mass ($d\mu > 0$) results in a shrinking orbit ($dR < 0$).

Multiplying both sides by R , we obtain a separable first-order differential equation:

$$\frac{dR}{R} = -C d\mu \quad (122)$$

We integrate this equation from the initial state of the simulation ($R_{\text{initial}}, \mu = 0$) to the final stalling state ($R_{\text{final}}, \mu = M_{\text{sp}}/M_1$):

$$\int_{R_{\text{initial}}}^{R_{\text{final}}} \frac{dR}{R} = -C \int_0^{M_{\text{sp}}/M_1} d\mu \quad (123)$$

$$\ln \left(\frac{R_{\text{final}}}{R_{\text{initial}}} \right) = -C \left(\frac{M_{\text{sp}}}{M_1} \right) \quad (124)$$

Exponentiating both sides, we arrive at the exact analytical solution for the stalling radius:

$$R_{\text{final}} = R_{\text{initial}} \exp \left[-C \left(\frac{M_{\text{sp}}}{M_1} \right) \right] \quad (125)$$

Since our N -body simulations consistently initialize at $R_{\text{initial}} = 1.0$ pc, the equation naturally simplifies to a pure exponential decay. Finally, we note that our numerical fit discovers a slight non-linearity in the exponent ($\alpha \approx 0.89$ rather than exactly 1). This physical deviation from the pure mathematical ideal arises because the dark matter spike is not uniform; it features a steep density profile. As the binary shrinks, it interacts with an increasingly dense core, causing a geometric stretch in the scattering efficiency that is well-parameterized by a stretched exponential $\exp(-C\mu^\alpha)$.

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